Volume Iii Reference

Contents

Structure of the Real Line

0.1 Real Line One Dimensional

0.2 Order Distance Length

Definition 0.2.1 (Distance on the Real Line)**.** The *distance* on $\mathbb{R}$ is the map $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ given by

$$d(x, y) = |x - y|.$$

Definition 0.2.2 (Length of a Bounded Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *length* of any bounded interval with endpoints a and b is

$$\ell = b - a.$$

Theorem 0.2.3 (The Distance Function Is a Metric)**.** *The map $d(x, y) = |x - y|$ satisfies, for all $x, y, z \in \mathbb{R}$: (i) $d(x, y) \geq 0$ with $d(x, y) = 0$ iff $x = y$; (ii) $d(x, y) = d(y, x)$; and (iii) $d(x, z) \leq d(x, y) + d(y, z)$. Hence $(\mathbb{R}, d)$ is a metric space.* *Go to proof.*

0.3 Absolute Value as Distance

Proposition 0.3.1 (Absolute Value Is Distance to the Origin)**.** *For every $a \in \mathbb{R}$, $|a| = d(a, 0)$.* *Go to proof.*

0.4 Intervals as Subsets

Definition 0.4.1 (Bounded Subset of the Real Line)**.** A set $E \subseteq \mathbb{R}$ is *bounded* exactly when there exists $M \geq 0$ such that $|x| \leq M$ for all $x \in E$; equivalently, E is contained in some closed interval $[-M, M]$.

Set Operations on Intervals

Definition 0.4.2 (Union of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *union* is the set

$$I \cup J = \{x \in \mathbb{R} : x \in I \text{ or } x \in J\}.$$

Definition 0.4.3 (Intersection of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *intersection* is the set

$$I \cap J = \{x \in \mathbb{R} : x \in I \text{ and } x \in J\}.$$

Definition 0.4.4 (Complement of an Interval)**.** Let $I \subseteq \mathbb{R}$ be an interval. Its *complement* in $\mathbb{R}$ is the set

$$I^c = \mathbb{R} \setminus I = \{x \in \mathbb{R} : x \notin I\}.$$

Definition 0.4.5 (Difference of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. The *difference* of I by J is the set

$$I \setminus J = \{x \in \mathbb{R} : x \in I \text{ and } x \notin J\} = I \cap J^c.$$

0.5 Neighborhoods and Balls

Definition 0.5.1 (Open Ball on the Real Line)**.** Let $x \in \mathbb{R}$ and $r > 0$. The *open ball* of radius r about x is

$$B_r(x) = \{\, y \in \mathbb{R} : |y - x| < r \,\} = (x - r,\, x + r).$$

Definition 0.5.2 (Neighborhood of a Point)**.** A set $N \subseteq \mathbb{R}$ is a *neighborhood* of $x \in \mathbb{R}$ exactly when there exists $r > 0$ with $B_r(x) \subseteq N$.

0.6 Open Sets

Definition 0.6.1 (Open Set in $\mathbb{R}$)**.** A set $U \subseteq \mathbb{R}$ is *open* exactly when

$$\forall x \in U\ \exists r > 0\ (B_r(x) \subseteq U).$$

Theorem 0.6.2 (Open Intervals Are Open)**.** *Every open interval (a, b), every open ray (a, ∞) and $(-\infty, b)$, and $\mathbb{R}$ itself are open sets.* *Go to proof.*

Theorem 0.6.3 (Unions and Finite Intersections of Open Sets)**.** *An arbitrary union of open subsets of $\mathbb{R}$ is open, and a finite intersection of open subsets of $\mathbb{R}$ is open.* *Go to proof.*

0.7 Closed Sets

Definition 0.7.1 (Closed Set in $\mathbb{R}$)**.** A set $F \subseteq \mathbb{R}$ is *closed* exactly when its complement $\mathbb{R} \setminus F$ is open.

Theorem 0.7.2 (Closed Sets Contain Their Limit Points)**.** *A set $F \subseteq \mathbb{R}$ is closed if and only if it contains every one of its limit points.* *Go to proof.*

0.8 Interior Exterior Boundary

Definition 0.8.1 (Interior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an interior point of E exactly when

$$\exists r > 0\ (B_r(x) \subseteq E).$$

Definition 0.8.2 (Exterior Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an exterior point of E exactly when

$$\exists r > 0\ (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Definition 0.8.3 (Boundary Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a boundary point of E exactly when

$$\forall r > 0\ (B_r(x) \cap E \neq \emptyset\ \wedge\ B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Definition 0.8.4 (Interior of a Set)**.** The *interior* of $E \subseteq \mathbb{R}$, written $\operatorname{int}(E)$, is the set of all interior points of E.

0.9 Limit and Isolated Points

Definition 0.9.1 (Limit Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a *limit point* of E exactly when

$$\forall r > 0\ \exists y \in E\ \big(0 < |y - x| < r\big).$$

Definition 0.9.2 (Isolated Point)**.** Let $E \subseteq \mathbb{R}$ and $x \in E$. Then x is an *isolated point* of E exactly when

$$\exists r > 0\ \forall y \in E\ \big(|y - x| < r \Rightarrow y = x\big).$$

0.10 Unions and Intersections

Theorem 0.10.1 (Closure Laws for Closed Sets). *An arbitrary intersection of closed subsets of* $\mathbb{R}$ *is closed, and a finite union of closed subsets of* $\mathbb{R}$ *is closed.* *Go to proof.*

0.11 Covers and Subcovers

Definition 0.11.1 (Open Cover). Let $E \subseteq \mathbb{R}$. An *open cover* of E is a family $\{U_i\}_{i\in I}$ of open subsets of $\mathbb{R}$ with

$$E \subseteq \bigcup_{i\in I} U_i.$$

Definition 0.11.2 (Subcover and Finite Subcover). Let $\{U_i\}_{i\in I}$ be an open cover of E. A *subcover* is a subfamily $\{U_i\}_{i\in J}$ with $J \subseteq I$ that still covers E. It is a *finite subcover* when J is finite.

0.12 Compactness on R

Definition 0.12.1 (Compact Set in $\mathbb{R}$). A set $K \subseteq \mathbb{R}$ is *compact* exactly when every open cover of K admits a finite subcover:

$$\forall \{U_i\}_{i\in I} \text{ open cover of } K \ \exists \text{finite } J \subseteq I \left(K \subseteq \bigcup_{i\in J} U_i \right).$$

Theorem 0.12.2 (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded). *If* $K \subseteq \mathbb{R}$ *is compact, then* K *is closed and bounded.* *Go to proof.*

0.13 Heine Borel

Theorem 0.13.1 (Closed Bounded Intervals Are Compact). *Every closed bounded interval* $[a, b] \subseteq \mathbb{R}$ *is compact.* *Go to proof.*

Theorem 0.13.2 (Heine–Borel Theorem for $\mathbb{R}$). *A set* $K \subseteq \mathbb{R}$ *is compact if and only if* K *is closed and bounded.* *Go to proof.*

Proofs

Bounds

0.14 Completeness Axiom

Axiom of Completeness

Definition 0.14.1 (Least Upper Bound Property)**.** Let $(S, \leq)$ be an ordered set. The set S has the *Least Upper Bound Property* if and only if every nonempty subset of S that is bounded above in S has a supremum in S.

0.14.1 Nested Intervals

Nested Interval Property

Theorem 0.14.2 (Nested Interval Property)**.** *Let (I_n) be a sequence of nonempty closed intervals in $\mathbb{R}$. If $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, then*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Go to proof.

0.14.2 Archimedean Property

Archimedean Property

Theorem 0.14.3 (Archimedean Property)**.** *For every real number x, there exists a natural number n such that $n > x$.*

Go to proof.

Corollary 0.14.4 (Archimedean Reciprocal Corollary)**.** *For every $x > 0$ and every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $nx > y$.*

Go to proof.

0.14.3 Density

Density

Definition 0.14.5 (Dense Subset)**.** Let $D \subseteq S \subseteq \mathbb{R}$. The set D is *dense in* S if and only if for every $x \in S$ and every $\varepsilon > 0$ there exists $d \in D$ such that $|x - d| < \varepsilon$.

Definition 0.14.6 (Order Dense Subset)**.** Let X be a linearly ordered set and let $A \subseteq X$. The subset A is *order dense in* X if and only if for every $a, b \in X$ with $a < b$ there exists $c \in A$ such that $a < c < b$.

Theorem 0.14.7 (Density of the Rational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.* *Go to proof.*

Theorem 0.14.8 (Density of the Irrational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$.* *Go to proof.*

Corollary 0.14.9 (Rational and Irrational Points in Every Open Interval)**.** *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number.* *Go to proof.*

0.14.4 Completeness Summary

0.15 Bounds and Extremal Values

0.15.1 Upper and Lower Bounds

Definition 0.15.1 (Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $b \in S$. A *bound* of A means a one-sided comparison between b and every element of A: either $x \leq b$ for all $x \in A$, or $b \leq x$ for all $x \in A$.

Definition 0.15.2 (Upper Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $u \in S$. The element u is an *upper bound* of A if every element of A is less than or equal to u.

Definition 0.15.3 (Lower Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $\ell \in S$. The element ℓ is a *lower bound* of A if every element of A is greater than or equal to ℓ.

Definition 0.15.4 (Bounded Above)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded above* if it has at least one upper bound in S.

Definition 0.15.5 (Bounded Below)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded below* if it has at least one lower bound in S.

Definition 0.15.6 (Bounded Set)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded* if it is bounded above and bounded below.

0.15.2 Suprema and Infima

Definition 0.15.7 (Supremum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $s \in S$. The element s is the *supremum* of A if s is an upper bound of A and no smaller element of S is an upper bound of A.

Definition 0.15.8 (Infimum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $i \in S$. The element i is the *infimum* of A if i is a lower bound of A and no larger element of S is a lower bound of A.

Proposition 0.15.9 (Uniqueness of the Supremum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s, t \in S$ are both suprema of A, then $s = t$.*
Go to proof.

Proposition 0.15.10 (Uniqueness of the Infimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i, j \in S$ are both infima of A, then $i = j$.*

Proposition 0.15.11 (Upper Bound Property of the Supremum)**.** *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $s = \sup A$ exists in S. Then every element of A lies below s:*

$$(\forall x \in A)(x \leq s).$$

Go to proof.

Proposition 0.15.12 (Lower Bound Property of the Infimum)**.** *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $i = \inf A$ exists in S. Then every element of A lies above i:*

$$(\forall x \in A)(i \leq x).$$

Lemma 0.15.13 (Subset Inclusion Preserves Upper Bounds). *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If u is an upper bound of B, then u is an upper bound of A.*
Go to proof.

Lemma 0.15.14 (Subset Inclusion Preserves Lower Bounds). *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If ℓ is a lower bound of B, then ℓ is a lower bound of A.*

Theorem 0.15.15 (Supremum Is Monotone Under Inclusion). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If $A \subseteq B$, then $\sup A \leq \sup B$.*
Go to proof.

Theorem 0.15.16 (Infimum Is Monotone Under Inclusion). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. If $A \subseteq B$, then $\inf B \leq \inf A$.*

Proposition 0.15.17 (Upper Bound Comparison with the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. For $u \in \mathbb{R}$, the number u is an upper bound of A if and only if $s \leq u$.*
Go to proof.

Proposition 0.15.18 (Lower Bound Comparison with the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. For $\ell \in \mathbb{R}$, the number ℓ is a lower bound of A if and only if $\ell \leq i$.*

Proposition 0.15.19 (Every Element Lies Below the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. Then $x \leq s$ for every $x \in A$.*
Go to proof.

Proposition 0.15.20 (Every Element Lies Above the Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. Then $i \leq x$ for every $x \in A$.*

Theorem 0.15.21 (Infimum Is Less Than or Equal to the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above. Then $\inf A \leq \sup A$.*
Go to proof.

Proposition 0.15.22 (Order Comparison of Suprema). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If every $x \in A$ is below some $y \in B$, then $\sup A \leq \sup B$.*
Go to proof.

Theorem 0.15.23 (Least Upper Bound Property Implies Existence of Suprema). *Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. By the least upper bound property of the real numbers, there exists a real number s such that $s = \sup S$.*

Theorem 0.15.24 (Greatest Lower Bound Property Implies Existence of Infima). *Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Then there exists a real number i such that $i = \inf S$.*

0.15.3 Maxima and Minima

Definition 0.15.25 (Maximum). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *maximum* of A if $m \in A$ and every element of A is less than or equal to m.

Definition 0.15.26 (Minimum). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *minimum* of A if $m \in A$ and every element of A is greater than or equal to m.

Proposition 0.15.27 (Uniqueness of the Maximum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both maxima of A, then $m = n$.*
Go to proof.

Proposition 0.15.28 (Uniqueness of the Minimum). *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both minima of A, then $m = n$.*

Proposition 0.15.29 (Maximum Implies Supremum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \max A$, then $m = \sup A$.*

Go to proof.

Proposition 0.15.30 (Minimum Implies Infimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \min A$, then $m = \inf A$.*

Proposition 0.15.31 (Supremum in the Set is the Maximum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s = \sup A$ and $s \in A$, then $s = \max A$.*

Go to proof.

Proposition 0.15.32 (Infimum in the Set is the Minimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i = \inf A$ and $i \in A$, then $i = \min A$.*

0.15.4 Epsilon Characterization

ε-Characterization of Supremum

Definition 0.15.33 (Epsilon Characterization of Supremum)**.** Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. The element s satisfies the *epsilon characterization of the supremum* of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $s - \varepsilon < a$.

Definition 0.15.34 (Epsilon Characterization of Infimum)**.** Let $A \subseteq \mathbb{R}$ and let $t \in \mathbb{R}$. The element t satisfies the *epsilon characterization of the infimum* of A if and only if t is a lower bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $a < t + \varepsilon$.

0.15.5 Extremal Bounds Summary

0.16 Algebra of Bounds

0.16.1 Set Operations and Bounds

Lemma 0.16.1 (Union Preserves Upper Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. If u_A is an upper bound of A and u_B is an upper bound of B, then $\max\{u_A, u_B\}$ is an upper bound of $A \cup B$. Go to proof.*

Lemma 0.16.2 (Union Preserves Lower Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. If ℓ_A is a lower bound of A and ℓ_B is a lower bound of B, then $\min\{\ell_A, \ell_B\}$ is a lower bound of $A \cup B$. Go to proof.*

Proposition 0.16.3 (Union Is Bounded Above Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded above if and only if A is bounded above and B is bounded above. Go to proof.*

Proposition 0.16.4 (Union Is Bounded Below Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded below if and only if A is bounded below and B is bounded below. Go to proof.*

Corollary 0.16.5 (Union Is Bounded Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded if and only if A is bounded and B is bounded. Go to proof.*

Lemma 0.16.6 (Subsets Preserve Upper Bounds)**.** *Let $C \subseteq A \subseteq \mathbb{R}$. If u is an upper bound of A, then u is an upper bound of C. Go to proof.*

Lemma 0.16.7 (Subsets Preserve Lower Bounds)**.** *Let $C \subseteq A \subseteq \mathbb{R}$. If ℓ is a lower bound of A, then ℓ is a lower bound of C. Go to proof.*

Corollary 0.16.8 (Intersections Inherit Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \cap B$, and every lower bound of A is a lower bound of $A \cap B$. Go to proof.*

Corollary 0.16.9 (Differences Inherit Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \setminus B$, and every lower bound of A is a lower bound of $A \setminus B$. Go to proof.*

Corollary 0.16.10 (Complements Inherit Ambient Bounds)**.** *Let $A \subseteq E \subseteq \mathbb{R}$, and take complements relative to E. Every upper bound of E is an upper bound of $E \setminus A$, and every lower bound of E is a lower bound of $E \setminus A$. Go to proof.*

0.16.2 Lattice Operations and Bounds

Definition 0.16.11 (Pointwise Maximum and Minimum Sets)**.** Let $A, B \subseteq \mathbb{R}$ be nonempty. Define

$$\max(A, B) := \{\max\{a, b\} : a \in A,\ b \in B\},$$

and

$$\min(A, B) := \{\min\{a, b\} : a \in A,\ b \in B\}.$$

Theorem 0.16.12 (Supremum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \max(A, B) = \max\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.16.13 (Infimum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \max(A, B) = \max\{\inf A, \inf B\}.$$

Go to proof.

Theorem 0.16.14 (Supremum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \min(A, B) = \min\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.16.15 (Infimum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \min(A, B) = \min\{\inf A, \inf B\}.$$

Go to proof.

0.16.3 Images, Inverse Images, and Extrema

Theorem 0.16.16 (Increasing Image Preserves Suprema)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be increasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\sup f(A) = f(\sup A), \qquad f(A) = \{f(a) : a \in A\}.$$

Go to proof.

Theorem 0.16.17 (Increasing Image Preserves Infima)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be increasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\inf f(A) = f(\inf A).$$

Go to proof.

Theorem 0.16.18 (Decreasing Image Sends Infimum to Supremum)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\sup f(A) = f(\inf A).$$

Go to proof.

Theorem 0.16.19 (Decreasing Image Sends Supremum to Infimum)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\inf f(A) = f(\sup A).$$

Go to proof.

Theorem 0.16.20 (Increasing Inverse Preserves Suprema)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

Theorem 0.16.21 (Increasing Inverse Preserves Infima)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.16.22 (Decreasing Inverse Sends Infimum to Supremum)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.16.23 (Decreasing Inverse Sends Supremum to Infimum)**.** *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

0.16.4 Dilation and Displacement

Definition 0.16.24 (Displacement, Dilation, and Reflection)**.** Let $A \subseteq \mathbb{R}$ and let $c, \lambda \in \mathbb{R}$. Define

$$A + c := \{a + c : a \in A\}, \qquad A - c := \{a - c : a \in A\}, \qquad \lambda A := \{\lambda a : a \in A\}.$$

The set $-A := (-1)A$ is the *reflection* of A through the origin.

Proposition 0.16.25 (Translation Preserves Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If u is an upper bound of A, then $u + c$ is an upper bound of $A + c$. Go to proof.*

Proposition 0.16.26 (Translation Preserves Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $\ell + c$ is a lower bound of $A + c$. Go to proof.*

Proposition 0.16.27 (Positive Dilation Preserves Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If u is an upper bound of A, then λu is an upper bound of λA. Go to proof.*

Proposition 0.16.28 (Positive Dilation Preserves Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If ℓ is a lower bound of A, then $\lambda \ell$ is a lower bound of λA. Go to proof.*

Proposition 0.16.29 (Negative Dilation Sends Lower Bounds to Upper Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If ℓ is a lower bound of A, then $\lambda \ell$ is an upper bound of λA. Go to proof.*

Proposition 0.16.30 (Negative Dilation Sends Upper Bounds to Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If u is an upper bound of A, then λu is a lower bound of λA. Go to proof.*

Corollary 0.16.31 (Reflection Swaps Upper and Lower Bounds)**.** *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $-\ell$ is an upper bound of $-A$. If u is an upper bound of A, then $-u$ is a lower bound of $-A$. Go to proof.*

0.16.5 Algebra of Suprema and Infima

Current Algebraic Properties of Supremum and Infimum

Definition 0.16.32 (Set Arithmetic Images)**.** Let $A, B \subseteq \mathbb{R}$. Define

$$A + B := \{a + b : a \in A,\ b \in B\}, \qquad A - B := \{a - b : a \in A,\ b \in B\},$$

$$AB := \{ab : a \in A,\ b \in B\}, \qquad A/B := \{a/b : a \in A,\ b \in B,\ b \neq 0\}.$$

Definition 0.16.33 (Reciprocal Image)**.** Let $A \subseteq \mathbb{R}$ and suppose $0 \notin A$. The reciprocal image of A is

$$A^{-1} := \{1/a : a \in A\}.$$

Theorem 0.16.34 (Translation Invariance of the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A + c) = \sup A + c, \qquad A + c = \{a + c : a \in A\}.$$

Go to proof.

Theorem 0.16.35 (Translation Invariance of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $c \in \mathbb{R}$. Then*

$$\inf(A + c) = \inf A + c.$$

Go to proof.

Theorem 0.16.36 (Scalar Multiplication by a Positive Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.16.37 (Positive Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k > 0$. Then*

$$\inf(kA) = k \inf A.$$

Go to proof.

Theorem 0.16.38 (Scalar Multiplication by a Negative Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.16.39 (Negative Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k < 0$. Then*

$$\inf(kA) = k \sup A.$$

Go to proof.

Theorem 0.16.40 (Negation Exchanges Supremum and Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below. Then $-A = \{-a : a \in A\}$ is bounded above and*

$$\sup(-A) = -\inf A.$$

Go to proof.

Theorem 0.16.41 (Supremum of a Sum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup(A + B) = \sup A + \sup B, \qquad A + B = \{a + b : a \in A,\ b \in B\}.$$

Go to proof.

Theorem 0.16.42 (Infimum of a Sum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf(A + B) = \inf A + \inf B.$$

Go to proof.

Theorem 0.16.43 (Supremum of a Difference Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \qquad A - B = \{a - b : a \in A,\ b \in B\}.$$

Go to proof.

Theorem 0.16.44 (Infimum of a Difference Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded below and B bounded above. Then*

$$\inf(A - B) = \inf A - \sup B.$$

Go to proof.

Theorem 0.16.45 (Supremum of a Dilation)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \qquad \lambda A = \{\lambda a : a \in A\}.$$

Go to proof.

Theorem 0.16.46 (Supremum of the Absolute-Value Image)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \qquad |A| = \{|a| : a \in A\}.$$

Go to proof.

Theorem 0.16.47 (Supremum of a Reciprocal Set)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\sup(A^{-1}) = \frac{1}{\inf A}.$$

Go to proof.

Theorem 0.16.48 (Infimum of a Reciprocal Set)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\inf(A^{-1}) = \frac{1}{\sup A}.$$

Go to proof.

Theorem 0.16.49 (Supremum of a Product Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup(AB) = \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.16.50 (Infimum of a Product Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf(AB) = \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.16.51 (Supremum of a Quotient Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\sup(A/B) = \max\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Theorem 0.16.52 (Infimum of a Quotient Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\inf(A/B) = \min\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Proposition 0.16.53 (Bounds for Suprema and Infima Under Set Addition)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \qquad \textit{and} \qquad \sup(A + B) \leq \sup A + \sup B.$$

Go to proof.

Proposition 0.16.54 (Upper Bounds Depend on the Ambient Order)**.** *Let $(S, \leq_S)$ and $(T, \leq_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order.* *Go to proof.*

Proposition 0.16.55 (Suprema Depend on the Ambient Set)**.** *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S, if it exists, need not agree with the supremum of A relative to S'.* *Go to proof.*

Theorem 0.16.56 (Ambient Existence of Supremum)**.** *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S.* *Go to proof.*

Lemma 0.16.57 (Rational Gap Example for Suprema)**.** *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$. *Go to proof.*

Proofs

Capstone

Functions

0.17 Algebra of Functions

Algebra of Functions

Proposition 0.17.1 (Composition of Injections Is Injective). *Let $f : A \to B$ and let $g : B \to C$. If f is injective and g is injective, then $g \circ f : A \to C$ is injective.*

Go to proof.

Proposition 0.17.2 (Composition of Surjections Is Surjective). Surjective$(f) \wedge$Surjective$(g) \Rightarrow$ Surjective$(g \circ f)$.

Go to proof.

Corollary 0.17.3 (Composition of Bijections Is Bijective). *Let $f : A \to B$ and let $g : B \to C$. If f is bijective and g is bijective, then $g \circ f : A \to C$ is bijective.*

Go to proof.

Proposition 0.17.4 (Inverse of a Bijection Is a Bijection). Bijective$(f) \Rightarrow$ Bijective(f^{-1}).

Go to proof.

Proposition 0.17.5 (Preimage Distributes over Union and Intersection). *Let $f : A \to B$, $S, T \subseteq B$. Then:*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

(iii) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

0.18 Real-Valued Functions

Real-Valued Functions

Definition 0.18.1 (Pointwise Sum of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise sum $f + g : X \to \mathbb{R}$ is the function defined by

$$(f + g)(x) = f(x) + g(x) \qquad (x \in X).$$

Definition 0.18.2 (Pointwise Difference of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise difference $f - g : X \to \mathbb{R}$ is the function defined by

$$(f - g)(x) = f(x) - g(x) \qquad (x \in X).$$

Definition 0.18.3 (Pointwise Product of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise product $fg : X \to \mathbb{R}$ is the function defined by

$$(fg)(x) = f(x)g(x) \qquad (x \in X).$$

Definition 0.18.4 (Pointwise Scalar Multiple of a Function)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $c \in \mathbb{R}$. The pointwise scalar multiple $cf : X \to \mathbb{R}$ is the function defined by

$$(cf)(x) = cf(x) \qquad (x \in X).$$

Definition 0.18.5 (Pointwise Absolute Value of a Function)**.** Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$. The pointwise absolute value $|f| : X \to \mathbb{R}$ is the function defined by

$$|f|(x) = |f(x)| \qquad (x \in X).$$

Definition 0.18.6 (Pointwise Maximum of Two Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise maximum $\max(f, g) : X \to \mathbb{R}$ is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \qquad (x \in X).$$

Definition 0.18.7 (Pointwise Quotient of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. Suppose $g(x) \neq 0$ for every $x \in X$. The pointwise quotient $f/g : X \to \mathbb{R}$ is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \qquad (x \in X).$$

Definition 0.18.8 (Linear Combination of Real-Valued Functions)**.** Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha f + \beta g : X \to \mathbb{R}$ is the function defined by

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) \qquad (x \in X).$$

Definition 0.18.9 (Real-Linear Rule)**.** Let $\mathcal{C}$ be a class of real-valued functions closed under real linear combinations. A rule T on $\mathcal{C}$ is real-linear if, whenever $f, g \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

whenever both sides are defined.

Definition 0.18.10 (Bounded Above Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded above on A if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \qquad (x \in A).$$

Definition 0.18.11 (Bounded Below Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded below on A if there exists $m \in \mathbb{R}$ such that

$$m \leq f(x) \qquad (x \in A).$$

Definition 0.18.12 (Bounded Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded on A if there exists $B > 0$ such that

$$|f(x)| \leq B \qquad (x \in A).$$

Definition 0.18.13 (Supremum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The supremum of f on A is

$$\sup_{x \in A} f(x) = \sup f(A),$$

provided the supremum of $f(A)$ exists.

Definition 0.18.14 (Infimum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The infimum of f on A is

$$\inf_{x \in A} f(x) = \inf f(A),$$

provided the infimum of $f(A)$ exists.

Definition 0.18.15 (Maximum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x^* \in A$ is a maximum point of f on A if

$$f(x) \leq f(x^*) \qquad (x \in A).$$

Definition 0.18.16 (Minimum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x_* \in A$ is a minimum point of f on A if

$$f(x_*) \leq f(x) \qquad (x \in A).$$

Definition 0.18.17 (Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is increasing on A if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.18 (Strictly Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly increasing on A if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.19 (Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is decreasing on A if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.20 (Strictly Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly decreasing on A if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.18.21 (Monotone Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is monotone on A if f is increasing on A or decreasing on A.

Definition 0.18.22 (Constant Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is constant on A if

$$f(x_1) = f(x_2) \qquad (x_1, x_2 \in A).$$

0.19 Subsets of $\mathbb{R}$

Subsets of $\mathbb{R}$

Definition 0.19.1 (ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **ε-neighbourhood** of x is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

Definition 0.19.2 (Deleted ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **deleted ε-neighbourhood** of x is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

Definition 0.19.3 (Cluster Point)**.** Let $X \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. The point x is a **cluster point** of X if

$$\forall \varepsilon > 0,\ \exists\, y \in X \setminus \{x\} :\ |y - x| < \varepsilon.$$

Theorem 0.19.4 (Sequential Characterization of Cluster Points)**.** *c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \to c$.*

Go to proof.

Definition 0.19.5 (Adherent Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **adherent point** of X if

$$\forall \varepsilon > 0 \; \exists y \in X \; (|y - x| < \varepsilon).$$

Definition 0.19.6 (Isolated Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in X$. The point x is an **isolated point** of X if

$$\exists \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X = \{x\}\big).$$

Definition 0.19.7 (Interior Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **interior point** of X if

$$\exists \varepsilon > 0 \; \big(V_\varepsilon(x) \subseteq X\big).$$

Definition 0.19.8 (Boundary Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is a **boundary point** of X if

$$\forall \varepsilon > 0 \; \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Definition 0.19.9 (Interior)**.** Let $X \subseteq \mathbb{R}$. The **interior** of X is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0 \; V_\varepsilon(x) \subseteq X\}.$$

Definition 0.19.10 (Boundary)**.** Let $X \subseteq \mathbb{R}$. The **boundary** of X is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0 \; (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

Definition 0.19.11 (Closure)**.**

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0 \; V_\varepsilon(x) \cap X \neq \emptyset\}.$$

Lemma 0.19.12 (Elementary Properties of Closures)**.** *Let $X, Y \subseteq \mathbb{R}$. Then: (i) $X \subseteq \overline{X}$; (ii) $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; (iii) $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; (iv) $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.*

Go to proof.

Definition 0.19.13 (Closed Subset of $\mathbb{R}$)**.** E is closed $\iff \overline{E} = E$.

Corollary 0.19.14 (Closed iff Sequentially Closed)**.** *X is closed $\iff \forall (a_n) \subseteq X, \; a_n \to x \Rightarrow x \in X$.*

Go to proof.

Corollary 0.19.15 (Every Point of an Interval Is a Cluster Point)**.** *If I is an interval, then every $x \in I$ is a cluster point of I.*

Go to proof.

Definition 0.19.16 (Bounded Subset of $\mathbb{R}$)**.** X is bounded $\iff \exists M > 0, \; \forall x \in X : |x| \leq M$.

Theorem 0.19.17 (Heine–Borel Theorem for $\mathbb{R}$)**.** *X is closed $\wedge$ X is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Go to proof.

Definition 0.19.18 (True Near a Point)**.** Property $P(f(x))$ is **true near** x_0 if $\exists \delta > 0, \; \forall x : 0 < |x - x_0| < \delta) \Rightarrow P(f(x))$.

Proofs

Discrete Calculus

0.20 Motivation

Motivation

Discrete Analogs of Calculus

Telescoping Phenomena

Why Discrete Calculus Matters

A Structural View

0.21 Foundations

0.21.1 Foundations

0.21.2 Summation Notation

Basic Properties of Summation

Splitting Sums

Changing the Index

Empty Sums

Nested Summations

Common Examples

0.21.3 The Binomial Theorem

Theorem 0.21.1 (Binomial Theorem)**.** *For any integer $n \geq 0$ and real numbers x, y,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^{k},$$

where the binomial coefficients *are*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Basic Properties of Binomial Coefficients

Connection with Discrete Calculus

Proposition 0.21.2 (Higher difference formula)**.** *For any function f,*

$$\Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k).$$

Go to proof.

Explicit Examples

0.22 Algebra of Summations

Linearity

Splitting a Sum

Index Renaming

Index Shifting

Constant Sum

Empty Sum

Nested Sums

Common Summation Identities

0.23 Difference Operators

0.23.1 Difference Operators

Motivation

Forward Difference

Definition 0.23.1 (Forward Difference Operator)**.** Let $f : \mathbb{Z} \to \mathbb{R}$. The *forward difference operator* Δ is defined by

$$\Delta f(n) = f(n+1) - f(n).$$

The function Δf measures the change in f when the input increases by one unit.

Example

0.23.2 Higher Differences

Definition 0.23.2 (Higher Differences)**.** The *second difference* of f is defined by

$$\Delta^2 f(n) = \Delta(\Delta f(n)).$$

More generally, the k-th difference $\Delta^k f(n)$ denotes the k-fold application of the difference operator, defined recursively by

$$\Delta^0 f = f, \qquad \Delta^k f = \Delta(\Delta^{k-1} f).$$

Example

0.23.3 Properties of the Difference Operator

Linearity

Proposition 0.23.3 (Linearity of Δ)**.** *For functions $f, g : \mathbb{Z} \to \mathbb{R}$ and constants $a, b \in \mathbb{R}$,*

$$\Delta(af + bg) = a\,\Delta f + b\,\Delta g.$$

Constant Rule

Proposition 0.23.4 (Constant rule)**.** *If c is a constant function, then $\Delta c = 0$.*

The Shift Operator and Operator Algebra

Definition 0.23.5 (Shift Operator)**.** The *shift operator E* and the *identity operator I* are defined by

$$Ef(n) = f(n+1), \qquad If(n) = f(n).$$

Theorem 0.23.6 ($\Delta = E - I$)**.**

$$\Delta = E - I.$$

0.23.4 Difference Tables and Polynomial Detection

Definition 0.23.7 (Difference Table)**.** Given a sequence $f(0), f(1), f(2), \ldots$, the *difference table* is constructed by listing the successive values of the function and then computing their forward differences $\Delta f(n) = f(n+1) - f(n)$, then $\Delta^2 f$, $\Delta^3 f$, and so on.

Example

Theorem 0.23.8 (Polynomial detection)**.** *Let $f(n)$ be a sequence generated by a polynomial of degree d. Then $\Delta^d f(n)$ is constant, and $\Delta^{d+1} f(n) = 0$.*

0.24 Telescoping Sums

0.24.1 Telescoping Sums

A Simple Example

General Form

The Telescoping Identity

Cancellation Pattern

Operator Form

0.25 Discrete Antiderivatives

0.25.1 Discrete Antiderivatives

Definition 0.25.1 (Discrete Antiderivative)**.** Let $f : \mathbb{Z} \to \mathbb{R}$ be a discrete function. A function $F : \mathbb{Z} \to \mathbb{R}$ is called a *discrete antiderivative* (or *discrete indefinite sum*) of f if

$$\Delta F(n) = f(n),$$

that is, $F(n+1) - F(n) = f(n)$ for all $n \in \mathbb{Z}$.

Examples

Constant of Integration

Proposition 0.25.2 (Constant of integration)**.** *If F is a discrete antiderivative of f, then so is $F + C$ for any constant $C \in \mathbb{R}$.*

Fundamental Theorem of Finite Calculus

0.26 Polynomial Basis

0.26.1 Polynomial Basis: Falling Factorials

The Difference Rule for Falling Factorials

Summary Table

Polynomial Representation in the Falling Factorial Basis

Consequences of the Difference Rule

Higher Differences of Falling Factorials

Connection to Binomial Coefficients

Worked Example: Expanding n^2

0.27 Calculus Rules

0.27.1 Calculus Rules

Product Rule

Summation by Parts

0.28 Expansions

0.28.1 Newton's Forward Difference Expansion

Structural Parallel with the Taylor Series

Proof Sketch

Reading Coefficients from the Difference Table

Worked Example: Expanding $f(n) = n^3$

Application: Computing $\sum_{k=0}^{n-1} k^2$

0.29 Special Functions

0.29.1 Special Functions in Discrete Calculus

The Discrete Exponential

Definition 0.29.1 (Discrete Exponential)**.** For a constant $a \neq -1$ and $n \in \mathbb{Z}$, the *discrete exponential* with base $(1 + a)$ is the function

$$E_a(n) = (1 + a)^n.$$

Proposition 0.29.2 (Discrete exponential rule)**.**

$$\Delta\,(1 + a)^n = a\,(1 + a)^n.$$

0.29.2 The Function 2^n in Discrete Calculus

Differencing 2^n

Antiderivative and Geometric Sum

Higher Differences of 2^n

Proposition 0.29.3 (Higher differences of 2^n)**.** *For all $k \geq 0$,*

$$\Delta^k \, 2^n = 2^n.$$

2^n in Combinatorics and the Newton Expansion

Discrete Differential Equation

Proposition 0.29.4 (Discrete IVP)**.** *The unique function $F : \mathbb{N} \to \mathbb{R}$ satisfying*

$$F(n+1) = 2\,F(n), \qquad F(0) = 1$$

is $F(n) = 2^n$.

Difference Formulas for Special Functions

Antiderivative Table

Proofs

Sequences

0.30 Sequence Definitions

0.30.1 Definitions

Definition 0.30.1 (Sequence)**.** Let X be a nonempty set. A **sequence** in X is a function

$$x : \mathbb{N} \to X.$$

The value $x(n)$ is written x_n and called the **n-th term** of the sequence. The sequence itself is denoted $(x_n)_{n\in\mathbb{N}}$, or simply (x_n) when the index set is clear.

Definition 0.30.2 (Real Sequence)**.** A **real sequence** is a sequence whose image is contained in $\mathbb{R}$. Equivalently, a real sequence is a function

$$x : \mathbb{N} \to \mathbb{R}.$$

0.30.2 Null and Constant Sequences

Null and Constant Sequences

Definition 0.30.3 (Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **constant** exactly when there exists $c \in \mathbb{R}$ such that $x_n = c$ for every $n \in \mathbb{N}$.

Definition 0.30.4 (Null Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **null sequence** exactly when for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for every $n \geq N$.

Theorem 0.30.5 (Constant Sequence Convergence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c.*

Go to proof.

Theorem 0.30.6 (Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.30.7 (Constant Null Sequence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Definition 0.30.8 (Ultimately Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **ultimately constant** exactly when there exist $N \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N$.

Theorem 0.30.9 (Difference from the Limit is Null)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*

Go to proof.

Theorem 0.30.10 (Ultimately Constant Sequence Convergence)**.** *Let (x_n) be a real sequence. If there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$, then (x_n) converges to c.*

Go to proof.

Theorem 0.30.11 (Constant Implies Ultimately Constant)**.** *Every constant real sequence is ultimately constant.*

Go to proof.

Theorem 0.30.12 (Ultimately Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.30.13 (Ultimately Constant Null Sequence)**.** *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Theorem 0.30.14 (Tail Equality Preserves Convergence)**.** *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L.*

Go to proof.

Definition 0.30.15 (Sequence Bounded Above)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded above** exactly when there exists $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$.

Definition 0.30.16 (Sequence Bounded Below)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded below** exactly when there exists $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$.

Definition 0.30.17 (Bounded Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded** exactly when there exists $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Theorem 0.30.18 (Eventually Bounded Above Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.19 (Eventually Bounded Below Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.20 (Eventually Bounded Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.30.21 (Bounded Sequence If and Only If Bounded Above and Below)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Go to proof.

Theorem 0.30.22 (Absolute Bound Gives Upper and Lower Bounds)**.** *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \leq K$ for every $n \in \mathbb{N}$, then $-K \leq x_n \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

Theorem 0.30.23 (Upper and Lower Bounds Give an Absolute Bound)**.** *Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

0.31 Examples and Counterexamples

Examples and Counterexamples

0.32 Convergence

Definition 0.32.1 (Convergence of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$. The sequence (x_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

The value L is called the **limit** of (x_n), written $x_n \to L$ or $\lim\limits_{n\to\infty} x_n = L$. A sequence that converges to some $L \in \mathbb{R}$ is called **convergent**; otherwise it is called **divergent**.

Definition 0.32.2 (Convergence — Neighborhood Formulation)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The sequence (x_n) **converges** to L if for every $\varepsilon > 0$ the ε-neighborhood

$$V_\varepsilon(L) \ :=\ (L - \varepsilon,\, L + \varepsilon)$$

contains all but finitely many terms of (x_n); that is,

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n \in V_\varepsilon(L)\big).$$

Theorem 0.32.3 (Equivalence of Convergence Formulations)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The following statements are equivalent.*

(a) $x_n \to L$.

(b) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(|x_n - L| < \varepsilon\big)$.

(c) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(L - \varepsilon < x_n < L + \varepsilon\big)$.

(d) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(x_n \in V_\varepsilon(L)\big)$.

Go to proof.

0.32.1 Limits

Theorem 0.32.4 (Uniqueness of Limits)**.** *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \to L$ and $x_n \to K$, then $L = K$.*

Go to proof.

Theorem 0.32.5 (Limit Preserves Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \to L$ and $y_n \to M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N_0$, then $L \leq M$.*

Go to proof.

Theorem 0.32.6 (Strict Limit Separation Gives Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.7 (Eventual Strict Comparison Preserves Weak Limit Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Theorem 0.32.8 (Constant Comparison for Sequence Limits)**.** *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n \leq B$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \geq N$, then $A \leq B$.

(c) If there exists $N \in \mathbb{N}$ such that $x_n \geq B$ for every $n \geq N$, then $A \geq B$.

(d) If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Theorem 0.32.9 (Constant Squeeze Theorem)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \leq x_n \leq L$ for every $n \geq N_0$, then $x_n \to L$.*

Go to proof.

Theorem 0.32.10 (Sequence Squeeze Theorem)**.** *Let (a_n), (x_n), and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \to L$ and $b_n \to L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \leq x_n \leq b_n$ for every $n \geq N_0$, then $x_n \to L$.*

Go to proof.

Theorem 0.32.11 (Absolute-Value Squeeze Theorem)**.** *Let (x_n) and (u_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $u_n \to 0$. If there exists $N_0 \in \mathbb{N}$ such that $|x_n - L| \leq u_n$ for every $n \geq N_0$, then $x_n \to L$.*

Go to proof.

0.32.2 Algebra of Limits

Algebra of Convergent Sequences

Definition 0.32.12 (Pointwise Sum of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise sum** of (x_n) and (y_n) is the real sequence $(x_n + y_n)$ defined by

$$(x + y)_n := x_n + y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.13 (Pointwise Difference of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise difference** of (x_n) and (y_n) is the real sequence $(x_n - y_n)$ defined by

$$(x - y)_n := x_n - y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.14 (Scalar Multiple of a Sequence)**.** Let (x_n) be a real sequence, and let $\alpha \in \mathbb{R}$. The **scalar multiple** $\alpha(x_n)$ is the real sequence (αx_n) defined by

$$(\alpha x)_n := \alpha x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.15 (Linear Combination of Real Sequences)**.** Let (x_n) and (y_n) be real sequences, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha(x_n) + \beta(y_n)$ is the real sequence $(\alpha x_n + \beta y_n)$ defined by

$$(\alpha x + \beta y)_n := \alpha x_n + \beta y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.16 (Pointwise Negation of a Sequence)**.** Let (x_n) be a real sequence. The **pointwise negation** of (x_n) is the real sequence $(-x_n)$ defined by

$$(-x)_n := -x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.17 (Pointwise Product of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise product** of (x_n) and (y_n) is the real sequence $(x_n y_n)$ defined by

$$(xy)_n := x_n y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.18 (Reciprocal Sequence)**.** Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$. The **reciprocal sequence** of (x_n) is the real sequence $(1/x_n)$ defined by

$$\left(\frac{1}{x}\right)_n := \frac{1}{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.19 (Pointwise Quotient of Sequences)**.** Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$. The **pointwise quotient** of (x_n) by (y_n) is the real sequence (x_n/y_n) defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.20 (Square Sequence)**.** Let (x_n) be a real sequence. The **square sequence** of (x_n) is the real sequence (x_n^2) defined by

$$(x^2)_n := x_n^2 \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.21 (Absolute Value Sequence)**.** Let (x_n) be a real sequence. The **absolute value sequence** of (x_n) is the real sequence $(|x_n|)$ defined by

$$(|x|)_n := |x_n| \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.32.22 (Square Root Sequence)**.** Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$. The **square root sequence** of (x_n) is the real sequence $(\sqrt{x_n})$ defined by

$$(\sqrt{x})_n := \sqrt{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.32.23 (Limit of a Scalar Multiple)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \to L$, then $\alpha x_n \to \alpha L$.*

Go to proof.

Theorem 0.32.24 (Limit of a Sum)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n + y_n \to L + M$.*

Go to proof.

Theorem 0.32.25 (Limit of a Negation)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $-x_n \to -L$.*

Go to proof.

Theorem 0.32.26 (Limit of a Difference)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n - y_n \to L - M$.*

Go to proof.

Theorem 0.32.27 (Limit of a Product)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n y_n \to LM$.*

Go to proof.

Theorem 0.32.28 (Nonzero Limit is Eventually Nonzero)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.29 (Limit of a Reciprocal)**.** *Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then $1/x_n \to 1/L$.*

Go to proof.

Theorem 0.32.30 (Limit of a Quotient)**.** *Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \to L$ and $y_n \to M$, then $x_n/y_n \to L/M$.*

Go to proof.

Theorem 0.32.31 (Limit of a Square)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $x_n^2 \to L^2$.*

Go to proof.

Theorem 0.32.32 (Limit of an Absolute Value)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $|x_n| \to |L|$.*

Go to proof.

Theorem 0.32.33 (Positive Limit is Eventually Positive)**.** *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.32.34 (Limit of a Square Root)**.** *Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \to L$, then $0 \leq L$ and $\sqrt{x_n} \to \sqrt{L}$.*

Go to proof.

Theorem 0.32.35 (Polynomial Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \to L$, then $p(x_n) \to p(L)$.*

Go to proof.

Theorem 0.32.36 (Rational Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \to L$, then*

$$\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}.$$

Go to proof.

0.33 Tails

Definition 0.33.1 (M-Tail of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $M \in \mathbb{N}$. The M**-tail** of (x_n) is the sequence

$$(x_n)_{n \geq M} \;:=\; \big(x_M,\, x_{M+1},\, x_{M+2},\, \ldots\big).$$

Theorem 0.33.2 (Convergence of a Tail)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m-tail $(x_{m+n})_{n \in \mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n \to \infty} x_{m+n} \;=\; \lim_{n \to \infty} x_n.$$

Go to proof.

Theorem 0.33.3 (Convergence by Domination)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim\limits_{n \to \infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \;\leq\; c\, a_n \qquad \forall n \geq m,$$

then $\lim\limits_{n \to \infty} x_n = L$.

Go to proof.

Theorem 0.33.4 (Ratio Limit Less Than One Implies Null)**.** *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n \to \infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n \to \infty} x_n = 0.$$

Go to proof.

0.34 Monotonicity

Monotonicity of Sequences

Definition 0.34.1 (Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **increasing** if

$$x_n \leq x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.2 (Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **decreasing** if

$$x_{n+1} \leq x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.3 (Strictly Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly increasing** if

$$x_n < x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.4 (Strictly Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **strictly decreasing** if

$$x_{n+1} < x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.34.5 (Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **monotone** if it is increasing or decreasing.

Definition 0.34.6 (Eventually Increasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually increasing** if there exists $N \in \mathbb{N}$ such that

$$x_n \leq x_{n+1} \qquad \text{for every } n \geq N.$$

Definition 0.34.7 (Eventually Decreasing Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually decreasing** if there exists $N \in \mathbb{N}$ such that

$$x_{n+1} \leq x_n \qquad \text{for every } n \geq N.$$

Definition 0.34.8 (Eventually Monotone Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **eventually monotone** if it is eventually increasing or eventually decreasing.

Theorem 0.34.9 (Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and bounded above, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If (x_n) is decreasing and bounded below, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.34.10 (Strict Monotonicity Implies Monotonicity)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is strictly increasing, then (x_n) is increasing.

(b) If (x_n) is strictly decreasing, then (x_n) is decreasing.

Go to proof.

Theorem 0.34.11 (Bounded Monotone Sequence Equivalences)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.

(b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.

(c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Go to proof.

Theorem 0.34.12 (Eventually Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.

(b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.

(c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Go to proof.

Theorem 0.34.13 (Unbounded Monotone Divergence)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and is not bounded above, then $x_n \to +\infty$.

(b) If (x_n) is decreasing and is not bounded below, then $x_n \to -\infty$.

Go to proof.

Theorem 0.34.14 (Algebraic Transformations Preserve Monotonicity)**.** *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

(a) If (x_n) is increasing, then $(x_n + c)$ is increasing.

(b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.

(c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.

(d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.

(e) If (x_n) is increasing, then $(-x_n)$ is decreasing.

(f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Go to proof.

0.35 Subsequences

Subsequences

Definition 0.35.1 (Strictly Increasing Index Map)**.** Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a function. The function σ is a **strictly increasing index map** if

$$k < \ell \Longrightarrow \sigma(k) < \sigma(\ell) \qquad \text{for all } k, \ell \in \mathbb{N}.$$

Definition 0.35.2 (Subsequence)**.** Let (x_n) be a real sequence. A **subsequence** of (x_n) is a sequence $(x_{\sigma(k)})_{k\in\mathbb{N}}$, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a strictly increasing index map.

Definition 0.35.3 (Subsequential Limit)**.** Let (x_n) be a real sequence and let $L \in \mathbb{R}$. The real number L is a **subsequential limit** of (x_n) if there exists a subsequence $(x_{\sigma(k)})$ such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Definition 0.35.4 (Has Convergent Subsequence)**.** Let (x_n) be a real sequence. The sequence (x_n) **has a convergent subsequence** if there exist a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ and a real number L such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Theorem 0.35.5 (Subsequence Indices Dominate the Identity)**.** *Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a strictly increasing index map. Then*

$$k \leq \sigma(k) \qquad \text{for every } k \in \mathbb{N}.$$

Go to proof.

Theorem 0.35.6 (Subsequences Preserve Limits)**.** *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then every subsequence of (x_n) converges to L.*

Go to proof.

Theorem 0.35.7 (Subsequential Limit of a Convergent Sequence)**.** *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then L is the only possible subsequential limit of (x_n).*

Go to proof.

Theorem 0.35.8 (Divergence by Two Subsequential Limits)**.** *Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.*

Go to proof.

Theorem 0.35.9 (Boundedness Passes to Subsequences)**.** *Every subsequence of a bounded real sequence is bounded.*

Go to proof.

Theorem 0.35.10 (Monotonicity Passes to Subsequences)**.** *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

Go to proof.

Theorem 0.35.11 (Subsequence of a Subsequence)**.** *Every subsequence of a subsequence of (x_n) is a subsequence of (x_n).*

Go to proof.

Theorem 0.35.12 (Eventual Properties Pass to Subsequences)**.** *Let $P(n)$ be a property of indices. If there exists $N \in \mathbb{N}$ such that $P(n)$ holds for every $n \geq N$, then for every strictly increasing index map σ there exists $K \in \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \geq K$.*

Go to proof.

Theorem 0.35.13 (Frequent Properties Yield Subsequences)**.** *Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \geq N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.*

Go to proof.

Theorem 0.35.14 (Subsequential Limits Respect Bounds)**.** *Let (x_n) be a real sequence and let L be a subsequential limit of (x_n). If $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then $m \leq L \leq M$.*

Go to proof.

Theorem 0.35.15 (Squeeze Passes to Subsequences)**.** *Let (a_n), (x_n), and (b_n) be real sequences. If $a_n \leq x_n \leq b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that*

$$a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)} \qquad \text{for every } k \geq K.$$

Go to proof.

Theorem 0.35.16 (Monotone Subsequence Theorem)**.** *Every real sequence has a monotone subsequence.*

Go to proof.

Theorem 0.35.17 (Bolzano-Weierstrass Theorem for Sequences)**.** *Every bounded real sequence has a convergent subsequence.*

Go to proof.

Theorem 0.35.18 (Sequential Compactness of Closed Bounded Intervals)**.** *Let $a, b \in \mathbb{R}$ with $a \le b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*

Go to proof.

0.36 Divergence

Divergence

Definition 0.36.1 (Divergent Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **divergent** if there is no $L \in \mathbb{R}$ such that $x_n \to L$.

Definition 0.36.2 (Divergence to Positive Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $+\infty$ if

$$\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \ge N\ \big(M < x_n\big).$$

Definition 0.36.3 (Divergence to Negative Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $-\infty$ if

$$\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \ge N\ \big(x_n < M\big).$$

Definition 0.36.4 (Oscillatory Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **oscillatory** if it is divergent, does not diverge to $+\infty$, and does not diverge to $-\infty$.

Theorem 0.36.5 (Divergence to Infinity Implies Real Divergence)**.** *Let (x_n) be a real sequence. If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) is divergent.*

Go to proof.

Theorem 0.36.6 (Two Subsequential Limits Force Divergence)**.** *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \ne K$, then (x_n) is divergent.*

Go to proof.

Theorem 0.36.7 (Unbounded Above Sequence Has a Positive-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to +\infty$.*

Go to proof.

Theorem 0.36.8 (Unbounded Below Sequence Has a Negative-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to -\infty$.*

Go to proof.

Theorem 0.36.9 (Bounded Divergence Produces Two Subsequential Limits)**.** *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \ne K$ such that L and K are subsequential limits of (x_n).*

Go to proof.

0.37 Liminf and Limsup

Liminf and Limsup

Definition 0.37.1 (Tail Supremum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail supremum sequence** of (x_n) is the real sequence (s_n) defined by

$$s_n := \sup\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.37.2 (Tail Infimum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail infimum sequence** of (x_n) is the real sequence (i_n) defined by

$$i_n := \inf\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.37.3 (Limit Superior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. The **limit superior** of (x_n) is

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} s_n.$$

Definition 0.37.4 (Limit Inferior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. The **limit inferior** of (x_n) is

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} i_n.$$

Theorem 0.37.5 (Tail Suprema Are Decreasing)**.** *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*

Go to proof.

Theorem 0.37.6 (Tail Infima Are Increasing)**.** *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*

Go to proof.

Theorem 0.37.7 (Liminf Is Below Limsup)**.** *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Go to proof.

Theorem 0.37.8 (Convergence iff Liminf Equals Limsup)**.** *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \to L$ if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Go to proof.

Theorem 0.37.9 (Limsup Is the Largest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n\to\infty} x_n$. Then S is a subsequential limit of (x_n), and every subsequential limit of (x_n) is less than or equal to S.*

Go to proof.

Theorem 0.37.10 (Liminf Is the Smallest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n\to\infty} x_n$. Then I is a subsequential limit of (x_n), and every subsequential limit of (x_n) is greater than or equal to I.*

Go to proof.

Theorem 0.37.11 (Oscillation Criterion via Liminf and Limsup)**.** *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Go to proof.

Theorem 0.37.12 (Limsup Comparison Under Eventual Order)**.** *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N$, then*

$$\limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} y_n.$$

Go to proof.

Theorem 0.37.13 (Liminf Comparison Under Eventual Order)**.** *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N$, then*

$$\liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n.$$

Go to proof.

Theorem 0.37.14 (Limsup Squeeze Under Eventual Order)**.** *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.$$

Go to proof.

Theorem 0.37.15 (Liminf Squeeze Under Eventual Order)**.** *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Go to proof.

0.38 Order-Theoretic Limit Constructions

Order-Theoretic Limit Constructions

Theorem 0.38.1 (Increasing Sequence Limit as Supremum)**.** *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.38.2 (Decreasing Sequence Limit as Infimum)**.** *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.38.3 (Tail Suprema and Infima Converge). *Let (x_n) be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \qquad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) and (i_n) converge.

Go to proof.

Theorem 0.38.4 (Bounded Sequence Has Limsup and Liminf). *Every bounded real sequence has a limit superior and a limit inferior.*

Go to proof.

0.39 Cluster Values of Sequences

Cluster Values of Sequences

Definition 0.39.1 (Cluster Value of a Sequence). Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The number L is a **cluster value** of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that

$$|x_n - L| < \varepsilon.$$

Theorem 0.39.2 (Cluster Values Are Subsequential Limits). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n).*

Go to proof.

Theorem 0.39.3 (Bounded Sequences Have Cluster Values). *Every bounded real sequence has at least one cluster value.*

Go to proof.

Theorem 0.39.4 (Limsup and Liminf Are Extremal Cluster Values). *Let (x_n) be a bounded real sequence. Then $\limsup_{n\to\infty} x_n$ is the largest cluster value of (x_n), and $\liminf_{n\to\infty} x_n$ is the smallest cluster value of (x_n).*

Go to proof.

0.40 Cauchy

Cauchy Sequences

Definition 0.40.1 (Cauchy Sequence). Let (x_n) be a real sequence. The sequence (x_n) is a **Cauchy sequence** if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall m, n \geq N \; \big(|x_m - x_n| < \varepsilon\big).$$

Theorem 0.40.2 (Convergent Sequences Are Cauchy). *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*

Go to proof.

Theorem 0.40.3 (Cauchy Sequences Are Bounded). *Every Cauchy real sequence is bounded.*

Go to proof.

Theorem 0.40.4 (Cauchy Sequence with a Convergent Subsequence). *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L.*

Go to proof.

Theorem 0.40.5 (Cauchy Criterion for Real Sequences). *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Go to proof.

Theorem 0.40.6 (Cauchy Criterion via Tails). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N-tail has pairwise term distances less than ε.*

Go to proof.

Theorem 0.40.7 (Cauchy Tail Diameter Criterion). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Go to proof.

Theorem 0.40.8 (Cauchy Sequences Have Vanishing Successive Differences). *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \to 0.$$

Go to proof.

Theorem 0.40.9 (Scalar Multiples of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*

Go to proof.

Theorem 0.40.10 (Sums of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n + y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.11 (Differences of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.12 (Linear Combinations of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.13 (Products of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.14 (Reciprocals of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*

Go to proof.

Theorem 0.40.15 (Quotients of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|y_n| \geq c$ for every $n \geq N_0$, then (x_n/y_n) is Cauchy.*

Go to proof.

Theorem 0.40.16 (Absolute Values of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*

Go to proof.

0.41 Applications of Sequences

Applications of Sequences

Definition 0.41.1 (Newton Sequence for $\sqrt{2}$)**.** The **Newton sequence for** $\sqrt{2}$ is the real sequence (x_n) defined by

$$x_1 := \frac{3}{2}, \qquad x_{n+1} := \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.2 (Newton Approximation of $\sqrt{2}$)**.** *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

Go to proof.

Definition 0.41.3 (Factorial Partial Sums)**.** The **factorial partial-sum sequence** is the real sequence (s_n) defined by

$$s_n := \sum_{k=0}^{n} \frac{1}{k!} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.4 (Factorial Partial Sums Approximate e)**.** *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e:*

$$\lim_{n\to\infty} s_n = e.$$

Go to proof.

Definition 0.41.5 (Compound-Interest Sequence)**.** The **compound-interest sequence** is the real sequence (c_n) defined by

$$c_n := \left(1 + \frac{1}{n}\right)^n \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.6 (Compound-Interest Approximation of e)**.** *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n\to\infty} c_n = e.$$

Go to proof.

Definition 0.41.7 (Decimal Truncation Sequence)**.** Let $\alpha \in \mathbb{R}$. The **decimal truncation sequence** of α is the real sequence (d_n) defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.41.8 (Decimal Truncations Converge)**.** *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α. Then*

$$\lim_{n\to\infty} d_n = \alpha.$$

Go to proof.

Proofs

Capstone

Series

0.42 Finite Series

Definition 0.42.1 (Finite Real Sequence on an Integer Interval)**.** Let $m, n \in \mathbb{Z}$ with $m \leq n$. A finite real sequence indexed from m to n is an assignment of a real number a_i to each integer i satisfying $m \leq i \leq n$. We denote such a sequence by $(a_i)_{i=m}^{n}$.

Definition 0.42.2 (Empty Finite Sum)**.** Let $m, n \in \mathbb{Z}$. If $n < m$, the finite sum from m to n is defined by

$$\sum_{i=m}^{n} a_i := 0.$$

Definition 0.42.3 (Finite Sum)**.** Let $m, n \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant integer indices. The notation

$$\sum_{i=m}^{n} a_i$$

denotes the finite sum, or finite series, of the terms indexed from m to n.

Definition 0.42.4 (Recursive Finite Sum)**.** Let $m \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant indices. The finite sum, also called a finite series, is defined recursively from the empty finite sum by the rule

$$\sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^{n} a_i \right) + a_{n+1} \qquad \text{whenever } n \geq m - 1.$$

Lemma 0.42.5 (Splitting a Finite Sum)**.** *Let $m \leq n < p$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq p$. Then*

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i.$$

Go to proof.

Definition 0.42.6 (Summation over a Finite Set)**.** Let X be a finite set with n elements, where $n \in \mathbb{N}$, and let $f : X \to \mathbb{R}$. Choose a bijection

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X.$$

The finite sum of f over X is defined by

$$\sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i)).$$

Proposition 0.42.7 (Finite Summations Are Well-Defined)**.** *Let X be a finite set with n elements, where $n \in \mathbb{N}$, let $f : X \to \mathbb{R}$, and let*

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X \qquad \textit{and} \qquad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X$$

be bijections. Then

$$\sum_{i=1}^{n} f(g(i)) = \sum_{i=1}^{n} f(h(i)).$$

Go to proof.

Proposition 0.42.8 (Empty Finite Set Summation)**.** *If X is empty, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

Go to proof.

Proposition 0.42.9 (Singleton Finite Set Summation)**.** *If X consists of a single element, $X = \{x_0\}$, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

Go to proof.

Proposition 0.42.10 (Substitution for Finite Set Summation)**.** *If X is a finite set, $f : X \to \mathbb{R}$ is a function, and $g : Y \to X$ is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

Go to proof.

Proposition 0.42.11 (Integer Interval as a Finite Set)**.** *Let $n \leq m$ be integers, and let*

$$X := \{i \in \mathbb{Z} : n \leq i \leq m\}.$$

If a_i is a real number assigned to each integer $i \in X$, then

$$\sum_{i=n}^{m} a_i = \sum_{i \in X} a_i.$$

Go to proof.

Proposition 0.42.12 (Disjoint Union Finite Set Summation)**.** *Let X and Y be disjoint finite sets, so $X \cap Y = \emptyset$, and let $f : X \cup Y \to \mathbb{R}$ be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left(\sum_{x \in X} f(x) \right) + \left(\sum_{y \in Y} f(y) \right).$$

Go to proof.

Proposition 0.42.13 (Addition Rule for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.42.14 (Scalar Multiplication Rule for Finite Set Summation)**.** *Let X be a finite set, let $f : X \to \mathbb{R}$ be a function, and let c be a real number. Then*

$$\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x).$$

Go to proof.

Proposition 0.42.15 (Monotonicity for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions such that $f(x) \leq g(x)$ for all $x \in X$. Then*

$$\sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.42.16 (Triangle Inequality for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ be a function. Then*

$$\left| \sum_{x \in X} f(x) \right| \leq \sum_{x \in X} |f(x)|.$$

Go to proof.

Lemma 0.42.17 (Iterated Summation over a Product)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x,y) \in X \times Y} f(x, y).$$

Go to proof.

Corollary 0.42.18 (Fubini's Theorem for Finite Series)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x,y) \in X \times Y} f(x, y) = \sum_{(y,x) \in Y \times X} f(x, y) = \sum_{y \in Y} \left(\sum_{x \in X} f(x, y) \right).$$

Go to proof.

Lemma 0.42.19 (Reindexing a Finite Sum)**.** *Let $m \leq n$ be integers, let k be an integer, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

Go to proof.

Lemma 0.42.20 (Finite Sum of Sums)**.** *Let $m \leq n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} (a_i + b_i) = \left(\sum_{i=m}^{n} a_i \right) + \left(\sum_{i=m}^{n} b_i \right).$$

Go to proof.

Lemma 0.42.21 (Finite Sum of Scalar Multiples)**.** *Let $m \le n$ be integers, let a_i be a real number assigned to each integer i satisfying $m \le i \le n$, and let $c \in \mathbb{R}$. Then*

$$\sum_{i=m}^{n}(ca_i) = c\left(\sum_{i=m}^{n} a_i\right).$$

Go to proof.

Lemma 0.42.22 (Triangle Inequality for Finite Series)**.** *Let $m \le n$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \le i \le n$. Then*

$$\left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n}|a_i|.$$

Go to proof.

Lemma 0.42.23 (Comparison Test for Finite Series)**.** *Let $m \le n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \le i \le n$. Suppose that $a_i \le b_i$ for all i satisfying $m \le i \le n$. Then*

$$\sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i.$$

Go to proof.

0.43 Infinite Series

Definition 0.43.1 (Formal Infinite Series)**.** A formal infinite series is an expression of the form

$$\sum_{n=m}^{\infty} a_n,$$

where m is an integer, and a_n is a real number for every integer $n \ge m$. This series is sometimes written as

$$a_m + a_{m+1} + a_{m+2} + \cdots.$$

Proofs

Function Sequences

Proofs

Elementary Functions

0.44 Hyperbolic Functions

Hyperbolic Functions

Definition 0.44.1 (Hyperbolic Functions)**.** For all $x \in \mathbb{R}$:

$$\cosh x \ := \ \frac{e^x + e^{-x}}{2}, \qquad \sinh x \ := \ \frac{e^x - e^{-x}}{2}, \qquad \tanh x \ := \ \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\coth x := \cosh x/\sinh x$.

Proposition 0.44.2 (Hyperbolic Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

Go to proof.

Proposition 0.44.3 (Addition Formulas for Hyperbolic Functions)**.** *For all $x, y \in \mathbb{R}$:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

Go to proof.

Proposition 0.44.4 (Limits of tanh)**.**

$$\lim_{x \to +\infty} \tanh x = 1, \qquad \lim_{x \to -\infty} \tanh x = -1.$$

Go to proof.

Definition 0.44.5 (Inverse Hyperbolic Functions)**.** Since sinh is strictly increasing and bijective $\mathbb{R} \to \mathbb{R}$, its inverse arcsinh is defined on all of $\mathbb{R}$. Since cosh is strictly increasing on $[0, \infty)$ with range $[1, \infty)$, its inverse arccosh is defined on $[1, \infty)$. Since tanh is strictly increasing with range $(-1, 1)$, arctanh is defined on $(-1, 1)$.
Explicit logarithmic formulas:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2+1}\right), \quad x \in \mathbb{R}.$$
$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2-1}\right), \quad x \geq 1.$$
$$\operatorname{arctanh} x = \tfrac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

0.45 The Logarithm

The Logarithm

Definition 0.45.1 (Natural Logarithm)**.** The **natural logarithm** $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x \;:=\; \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x \;:=\; \int_1^x \frac{1}{t}\, dt.$$

Proposition 0.45.2 (Fundamental Logarithmic Limit)**.**

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$$

Go to proof.

Proposition 0.45.3 (Every Power Dominates the Logarithm)**.** *For every $r > 0$:*

$$\lim_{x \to \infty} \frac{\ln x}{x^r} = 0 \qquad \textit{and} \qquad \lim_{x \to 0^+} x^r \ln x = 0.$$

Go to proof.

0.46 Power Functions and the Exponential

Power Functions and the Exponential

Definition 0.46.1 (Power Function)**.** Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r \;:=\; e^{r \ln x}.$$

Definition 0.46.2 (The Number e)**.** The **Euler number** is

$$e \;:=\; \sum_{n=0}^{\infty} \frac{1}{n!} \;=\; \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n \;\approx\; 2.71828\ldots$$

Both expressions define the same real number.

Definition 0.46.3 (Exponential Function)**.** The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is defined by the absolutely convergent power series

$$e^x \;:=\; \exp(x) \;:=\; \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in \mathbb{R}.$$

Proposition 0.46.4 (Fundamental Exponential Limit)**.**

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

Go to proof.

Proposition 0.46.5 (Compound Interest Limit)**.**

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e.$$

Go to proof.

Proposition 0.46.6 (Exponential Dominates Every Power)**.** *For every $r > 0$:*

$$\lim_{x\to\infty}\frac{x^r}{e^x} = 0.$$

Go to proof.

0.47 Trigonometric Functions

Trigonometric Functions

Definition 0.47.1 (Sine and Cosine via Power Series)**.** The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\sin x \;:=\; \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!} \;=\; x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x \;:=\; \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{(2n)!} \;=\; 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Proposition 0.47.2 (Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\sin^2 x + \cos^2 x = 1.$$

Go to proof.

Proposition 0.47.3 (Addition Formulas)**.** *For all $x, y \in \mathbb{R}$:*

$$\sin(x+y) = \sin x\cos y + \cos x \sin y, \qquad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Go to proof.

Definition 0.47.4 (Tangent, Cotangent, Secant, Cosecant)**.**

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

Definition 0.47.5 (Inverse Trigonometric Functions)**.**

- $\arcsin : [-1,1] \to [-\pi/2, \pi/2]$ is the inverse of $\sin\big|_{[-\pi/2,\pi/2]}$.
- $\arccos : [-1,1] \to [0,\pi]$ is the inverse of $\cos\big|_{[0,\pi]}$.
- $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is the inverse of $\tan\big|_{(-\pi/2,\pi/2)}$.

Theorem 0.47.6 (Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0}\frac{\sin x}{x} = 1.$$

Go to proof.

Proposition 0.47.7 (Second Fundamental Trigonometric Limit)**.**

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

Go to proof.

Proofs

Continuity

0.48 Limits

Limits of Functions

Definition 0.48.1 (ε-Closeness of a Function to a Value)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function f is **ε-close to L** if $|f(x) - L| \leq \varepsilon$ for all $x \in X$.

Definition 0.48.2 (Local δ-Closeness Near a Point)**.** Let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, let x_0 be adherent to X, and let $\varepsilon, \delta > 0$. The function f is **ε-close to L near x_0 within δ** if

$$\forall x \in X \ \big(|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon\big).$$

Definition 0.48.3 (Limit of a Function)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. A real number L is the **limit of f at** c, written $\lim_{x\to c} f(x) = L$, if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Go to uniqueness proof.

Definition 0.48.4 (Sequential Definition of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, the sequence $(f(x_n))$ converges to L.

Definition 0.48.5 (Neighbourhood Form of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **neighbourhood sense** holds if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a deleted δ-neighbourhood $V^*_\delta(c)$ such that

$$x \in V^*_\delta(c) \cap A \quad \Longrightarrow \quad f(x) \in V_\varepsilon(L).$$

Proposition 0.48.6 (Neighbourhood Form of the Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a δ-neighbourhood $V_\delta(c)$ such that $x \in V^*_\delta(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Go to proof.

Theorem 0.48.7 (Uniqueness of Limits of Functions)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then f has at most one limit at c.*

Go to proof.

Proposition 0.48.8 (Sequential Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Corollary 0.48.9 (Three Equivalent Forms of the Function Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The ε-δ form of $\lim_{x\to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

Go to proof.

Theorem 0.48.10 (Limit Implies Local Boundedness). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \le M$ for all $x \in A$ with $0 < |x - c| < \delta$.*

Go to proof.

Theorem 0.48.11 (Bounded Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $a, b \in \mathbb{R}$. If $a \le f(x) \le b$ for all $x \in A$ with $x \ne c$, and $\lim_{x\to c} f(x) = L$, then $a \le L \le b$.*

Go to proof.

Theorem 0.48.12 (Squeeze Theorem for Function Limits). *Let $f, g, h : A \to \mathbb{R}$ and let c be a cluster point of A. Suppose $\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L$, and suppose there exists $\delta_0 > 0$ such that*

$$\forall x \in A,\ 0 < |x - c| < \delta_0 \ \Rightarrow\ f(x) \le h(x) \le g(x).$$

Then $\lim_{x\to c} h(x) = L$.

Go to proof.

Theorem 0.48.13 (Limits Are Local). *Let $E \subseteq X \subseteq \mathbb{R}$, let x_0 be a cluster point of E, let $f : X \to \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta, x_0+\delta)} f(x) = L.$$

Go to proof.

Proposition 0.48.14 (Limit of Absolute Value). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, then $\lim_{x\to c} |f(x)| = |L|$.*

Go to proof.

Theorem 0.48.15 (Composition of Limits). *Let $f : A \to B$, let $g : B \to \mathbb{R}$, let c_1 be a cluster point of A, and let $c_2 \in \mathbb{R}$ be a cluster point of B. Suppose $\lim_{x\to c_1} f(x) = c_2$ and $\lim_{y\to c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x\to c_1} g(f(x)) = L$.*

Go to proof.

Theorem 0.48.16 (Monotone Limit Theorem). *Let $a < b$ and let $f : (a, b) \to \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing f and every $x_0 \in (a, b)$:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

Go to proof.

Proposition 0.48.17 (Cauchy Criterion for Function Limits). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall\ \text{OutputTolerance}(\varepsilon),\ \exists\ \text{InputTolerance}(\delta),\ \forall x, y \in A:\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta \ \Rightarrow\ |f(x)-f(y)| < \varepsilon.$$

Go to proof.

Algebra of Limits

Theorem 0.48.18 (Sum Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f + g)(x) = L + M.$$

Go to proof.

Theorem 0.48.19 (Difference Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f - g)(x) = L - M.$$

Go to proof.

Theorem 0.48.20 (Scalar Multiple Rule for Function Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then*

$$\lim_{x\to c}(\alpha f)(x) = \alpha L.$$

Go to proof.

Theorem 0.48.21 (Product Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(fg)(x) = LM.$$

Go to proof.

Theorem 0.48.22 (Quotient Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Go to proof.

One-Sided Limits

Definition 0.48.23 (Right-Hand Limit). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. A real number L is the **right-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Definition 0.48.24 (Left-Hand Limit). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. A real number L is the **left-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Theorem 0.48.25 (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x\to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Corollary 0.48.26 (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x\to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Theorem 0.48.27 (Two-Sided Limit via One-Sided Limits). *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x\to c} f(x) = L$ if and only if both $\lim_{x\to c^+} f(x) = L$ and $\lim_{x\to c^-} f(x) = L$.*

Go to proof.

0.49 Point Continuity

Continuity

Definition 0.49.1 (Relative Neighborhood)**.** Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of a in E if and only if there exists $\delta > 0$ such that $U = E \cap (a - \delta, a + \delta)$. Equivalently, every neighborhood of a in E is a set of the form $U_E(a) := E \cap (a - \delta, a + \delta)$ for some $\delta > 0$.

Definition 0.49.2 (Continuity at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is **continuous at** c if and only if

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A : \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Definition 0.49.3 (Continuity at a Point — Neighbourhood Form)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is continuous at c if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

Definition 0.49.4 (Continuity at a Point — Sequential Form)**.** Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f \colon A \to \mathbb{R}$. The function f is continuous at c if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in A, the convergence $\lim_{n\to\infty} x_n = c$ implies $\lim_{n\to\infty} f(x_n) = f(c)$.

Characterization of Discontinuity

Definition 0.49.5 (Point of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f \colon A \to \mathbb{R}$, and let $a \in A$. The function f is discontinuous at a if and only if f is not continuous at a.

Definition 0.49.6 (Sequential Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq A$ such that $x_n \to c$ while $f(x_n) \not\to f(c)$.

Definition 0.49.7 (Neighbourhood Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f \colon A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a neighborhood V of $f(c)$ such that for every neighborhood U of c relative to A there exists $x \in U \cap A$ with $f(x) \notin V$.

Lemma 0.49.8 (Equivalent Formulations of Discontinuity at a Point)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The following are equivalent:*

1. *c is a point of discontinuity in the ε-δ sense.*
2. *c is a sequential point of discontinuity.*
3. *c is a neighbourhood point of discontinuity.*

Go to proof.

Definition 0.49.9 (Types of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $x_0 \in A$ with f discontinuous at x_0.

(i) *Removable* iff $\exists L \in \mathbb{R} : \lim_{x\to x_0} f(x) = L$ and $L \neq f(x_0)$.

(ii) *Jump* iff $\exists L_-, L_+ \in \mathbb{R} : \lim_{x\to x_0^-} f(x) = L_-$, $\lim_{x\to x_0^+} f(x) = L_+$, and $L_- \neq L_+$.

(iii) *Essential* iff it is not removable.

Oscillation

Definition 0.49.10 (Oscillation on a Set)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $E \subseteq A$. The oscillation of f on E is the extended real number

$$\omega(f; E) \coloneqq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\} = \operatorname{diam}(f(E)).$$

Definition 0.49.11 (Oscillation at a Point)**.** Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \varnothing$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of f at x_0 is the extended real number

$$\omega(f; x_0) = \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big),$$

with $\omega(f; A)$ denoting the oscillation of f on the set A. Because the function $\delta \mapsto \omega\big(f;\, U_\delta(x_0) \cap E\big)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

Theorem 0.49.12 (Continuity Characterised by Oscillation)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f; c) = 0$. Go to proof.*

Proposition 0.49.13 (Discontinuity Set via Oscillation)**.** *Let $f : E \to \mathbb{R}$. Then*

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}.$$

Go to proof.

0.50 Global Theorems

Intermediate Value Theorem

Definition 0.50.1 (Bounded on a Set)**.** Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The function f is bounded on A if and only if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.

Theorem 0.50.2 (Boundedness Theorem)**.** *Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a, b]$. Go to proof.*

Definition 0.50.3 (Absolute Maximum and Absolute Minimum)**.** Let $A \subseteq \mathbb{R}$ and let $f\colon A \to \mathbb{R}$. A point $x^* \in A$ is an absolute maximum of f on A if and only if $f(x^*) \geq f(x)$ for every $x \in A$. A point $x_* \in A$ is an absolute minimum of f on A if and only if $f(x_*) \leq f(x)$ for every $x \in A$.

Theorem 0.50.4 (Extreme Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a, b]$. Go to proof.*

Theorem 0.50.5 (Location of Roots)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a, b)$ such that $f(c) = 0$. Go to proof.*

Theorem 0.50.6 (Bolzano's Intermediate Value Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f\colon I \to \mathbb{R}$ be continuous on I, and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that $\min\{a, b\} < c < \max\{a, b\}$ and $f(c) = k$. Go to proof.*

Theorem 0.50.7 (Image of Closed Bounded Interval Is Closed Bounded Interval)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then $f([a, b]) = [\min_{[a,b]} f, \max_{[a,b]} f]$. Go to proof.*

Theorem 0.50.8 (Preservation of Intervals)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous. Then $f(I)$ is an interval. Go to proof.*

Theorem 0.50.9 (Darboux Property)**.** *Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$ or $f(x) < c$ for all $x \in [a, b]$. Go to proof.*

0.51 Uniform Continuity

Uniform Continuity

Definition 0.51.1 (Uniform Continuity)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is uniformly continuous on A if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

Theorem 0.51.2 (Algebra of Uniform Continuity on a General Domain)**.** *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on A, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A.*

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product fg is not uniformly continuous on A.

Go to proof.

Theorem 0.51.3 (Algebra of Uniform Continuity on a Bounded Domain)**.** *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A.

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A.

Go to proof.

Proposition 0.51.4 (Sequential Characterization of Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. Then f is uniformly continuous on A if and only if for every pair of sequences (x_n) and (y_n) in A with $\lim_{n\to\infty}(x_n - y_n) = 0$ we have $\lim_{n\to\infty}(f(x_n) - f(y_n)) = 0$.* *Go to proof.*

Theorem 0.51.5 (Heine–Cantor Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is uniformly continuous on $[a, b]$.* *Go to proof.*

Proposition 0.51.6 (Uniform Continuity Preserves Cauchy Sequences)**.** *Let $S \subseteq \mathbb{R}$, let $f : S \to \mathbb{R}$, and let (x_n) be a sequence in S. If f is uniformly continuous on S and (x_n) is a Cauchy sequence in S, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.* *Go to proof.*

Definition 0.51.7 (Lipschitz Condition)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $K > 0$. The function f satisfies the Lipschitz condition on A with constant K if and only if $|f(x) - f(u)| \leq K|x - u|$ for every $x, u \in A$.

Proposition 0.51.8 (Lipschitz Implies Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then f is uniformly continuous on A.* *Go to proof.*

Theorem 0.51.9 (Algebra of Lipschitz Functions)**.** *Let $A \subseteq \mathbb{R}$ and let $f, g : A \to \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.*

(a) For every $c \in \mathbb{R}$, the function cf is Lipschitz on A with constant $|c|K_f$.

(b) The functions $f + g$ and $f - g$ are Lipschitz on A with constant $K_f + K_g$.

(c) If $B \subseteq \mathbb{R}$, $h : B \to \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$.

(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then fg is Lipschitz on A with constant $M_g K_f + M_f K_g$.

(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if g is Lipschitz on A with constant K_g, then f/g is Lipschitz on A with constant $K_f/k + M_f K_g/k^2$.

Go to proof.

Definition 0.51.10 (Bi-Lipschitz Map)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The map f is bi-Lipschitz on A if and only if there exist constants $0 < \alpha \le K$ such that for every $x, u \in A$,

$$\alpha|x - u| \le |f(x) - f(u)| \le K|x - u|.$$

Any such pair (α, K) is called a bi-Lipschitz constant pair for f on A.

Theorem 0.51.11 (Inverse of a Bi-Lipschitz Map Is Lipschitz)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x - y| \le |f(x) - f(y)| \le K|x - y|$ for all $x, y \in A$. Then f is injective, the inverse mapping $f^{-1} : f(A) \to A$ is well defined, and*

$$\frac{1}{K}|u - v| \le |f^{-1}(u) - f^{-1}(v)| \le \frac{1}{\alpha}|u - v| \quad \textit{for all } u, v \in f(A).$$

Go to proof.

0.52 Monotone Functions

Monotone Functions and Continuity

Theorem 0.52.1 (One-Sided Limits of Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let c be interior to I, and let $f : I \to \mathbb{R}$ be increasing. Define*

$$L_- := \sup\{f(x) : x \in I,\ x < c\}, \qquad L_+ := \inf\{f(x) : x \in I,\ c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \to c^-} f(x) = L_-, \qquad \lim_{x \to c^+} f(x) = L_+.$$

Go to proof.

Corollary 0.52.2 (Continuity Criterion for Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone increasing, and let c be an interior point of I. Then f is continuous at c if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x \to c^-} f(x)$ and $f(c^+) = \lim_{x \to c^+} f(x)$.* *Go to proof.*

Definition 0.52.3 (Jump of a Function)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be increasing, and let c be an interior point of I. If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of f at c is

$$j_f(c) := f(c^+) - f(c^-).$$

Proposition 0.52.4 (Discontinuities of a Monotone Function Are of First Kind)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a. If f is discontinuous at a, then both one-sided limits $\lim_{x \uparrow a} f(x)$ and $\lim_{x \downarrow a} f(x)$ exist in $\mathbb{R}$.* *Go to proof.*

Corollary 0.52.5 (Jump Intervals Are Disjoint)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of f, then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint.**Go to proof.*

Theorem 0.52.6 (Discontinuities of a Monotone Function Are Countable)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable.* *Go to proof.*

Corollary 0.52.7 (Monotone Function Is Continuous on a Dense Set)**.** *Let $f : [a, b] \to \mathbb{R}$ be monotone. Then the set $\{x \in [a, b] : f \text{ is continuous at } x\}$ is dense in $[a, b]$.* *Go to proof.*

Proposition 0.52.8 (Continuous Injective Is Strictly Monotone)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is injective if and only if f is strictly monotone on $[a, b]$.* *Go to proof.*

Theorem 0.52.9 (Continuous Inverse Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Then the inverse function $f^{-1} : f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$.* *Go to proof.*

Limits Superior and Inferior of Functions

Definition 0.52.10 (Limits Superior and Inferior of a Function)**.** Let $f : E \to \mathbb{R}$ and let x_0 be an accumulation point of E. For each $\delta > 0$ set

$$U(\delta) = \{\, f(x) : x \in E,\ 0 < |x - x_0| < \delta \,\}.$$

The extended real numbers $\limsup_{x\to x_0} f(x)$ and $\liminf_{x\to x_0} f(x)$ are defined by

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta), \qquad \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

Proposition 0.52.11 ($\limsup \geq \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let x_0 be a limit point of A, and let $f : A \to \mathbb{R}$. Then*

$$\limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

Go to proof.

Proposition 0.52.12 (Limit Exists iff $\limsup = \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals L if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$.* *Go to proof.*

0.53 Limits at Infinity

Limits at Infinity

Definition 0.53.1 (Infinite Adherent Points)**.** Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

Definition 0.53.2 (Limit at Infinity)**.** Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of f at $+\infty$ equals L if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

0.54 Approximation

Approximation of Continuous Functions

Theorem 0.54.1 (Step Function Approximation)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a, b] \to \mathbb{R}$ such that*

$$\sup_{x\in[a,b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

Go to proof.

Definition 0.54.2 (Piecewise Linear Function)**.** Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a, b] \to \mathbb{R}$. The function g is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \cdots < x_m = b$ such that $g|_{[x_{k-1},x_k]}$ is affine for every $k \in \{1, \ldots, m\}$.

Theorem 0.54.3 (Piecewise Linear Approximation)**.** *Let $f\colon [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon\colon [a, b] \to \mathbb{R}$ such that $\sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$.* *Go to proof.*

Theorem 0.54.4 (Weierstrass Approximation Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$. Go to proof.*

Definition 0.54.5 (Bernstein Polynomial)**.** Let $n \in \mathbb{N}$ and let $f : [0, 1] \to \mathbb{R}$. A function $B_n : [0, 1] \to \mathbb{R}$ is called the nth Bernstein polynomial of f if and only if, for every $x \in [0, 1]$,

$$B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right)\binom{n}{k} x^k (1 - x)^{n-k}.$$

Theorem 0.54.6 (Bernstein Approximation Theorem)**.** *Let $f : [0, 1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n denote the nth Bernstein polynomial of f. Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$. Go to proof.*

0.55 Gauge

Gauges and Tagged Partitions

Definition 0.55.1 (Partition)**.** Let $a < b$ and set $I = [a, b]$. A finite ordered set $P = \{x_0, \dots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of I if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \cdots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \dots, n$ subdivide I.

Definition 0.55.2 (Tagged Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a tagged partition of $[a, b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ of $[a, b]$ with $a = x_0 < x_1 < \cdots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \dots, n$. Equivalently, a tagged partition is exactly a partition P of $[a, b]$ together with one tag chosen inside every subinterval of P.

Definition 0.55.3 (Gauge)**.** Let $a, b \in \mathbb{R}$ with $a < b$ and set $I = [a, b]$. A gauge on I is a function $\delta : I \to (0, \infty)$.

Definition 0.55.4 (δ-Fine Partition)**.** Let $[a, b]$ be a closed interval, let $\delta : [a, b] \to (0, \infty)$ be a gauge, and let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$. The tagged partition $\dot{\mathcal{P}}$ is δ-fine if and only if for every $i \in \{1, \dots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

Lemma 0.55.5 (Every Point Is Covered by a Tag)**.** *Let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ that is δ-fine on $[a, b]$, and let $x \in [a, b]$. Then there exists $i \in \{1, \dots, n\}$ such that $|x - t_i| < \delta(t_i)$. Go to proof.*

Theorem 0.55.6 (Cousin's Theorem)**.** *Every gauge $\delta : [a, b] \to (0, \infty)$ on a closed interval $[a, b]$ admits a δ-fine tagged partition of $[a, b]$. Go to proof.*

Tagged Partitions and Mesh

Definition 0.55.7 (Mesh of a Partition)**.** Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. A real number m is the *mesh* (or *norm*) of P if and only if

$$m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

The mesh of P is denoted by $\|P\|$ and is the unique m satisfying this equivalence.

Definition 0.55.8 (Refinement)**.** Let $[a, b] \subseteq \mathbb{R}$ and let P and Q be partitions of $[a, b]$. Then Q is a refinement of P if and only if every point of P is also a point of Q; equivalently, $P \subseteq Q$.

Definition 0.55.9 (Common Refinement)**.** Let $a < b$ and let P and Q be partitions of $[a, b]$. Their common refinement is the partition R obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a, b]$.

Proofs

Differentiation

0.56 The Tangent Problem

Definition 0.56.1 (Secant Line)**.** Let $f : A \to \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Definition 0.56.2 (Difference Quotient)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and let $x \in A$ with $x \neq c$. The *difference quotient* of f from c to x is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

Definition 0.56.3 (Tangent Line)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c. The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

0.57 The Derivative

Definition 0.57.1 (Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ be a real-valued function and $c \in \text{int}(D)$. We say f is *differentiable at* c with *derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Definition 0.57.2 (Topological Definition of Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ and $c \in \text{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L, there exists a neighbourhood U of c such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

Definition 0.57.3 (Sequential Definition of Derivative at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. We say that f is differentiable at c with derivative L if

$$\forall (x_n) \subseteq A \setminus \{c\}\ \big(x_n \to c \Rightarrow \tfrac{f(x_n) - f(c)}{x_n - c} \to L \big).$$

Theorem 0.57.4 (Equivalence of Definitions of the Derivative at a Point)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$ be a limit point of A. The following are equivalent:*

1. *f is differentiable at c in the ε-δ sense with derivative L.*
2. *f is differentiable at c in the topological sense with derivative L.*
3. *f is differentiable at c in the sequential sense with derivative L.*

Go to proof.

Proposition 0.57.5 (Derivative: h-Form Equivalence)**.** *Let $f : A \to \mathbb{R}$ and $c \in \text{int}(A)$. Then f is differentiable at c with derivative L if and only if*

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

0.57.1 First Consequences

Theorem 0.57.6 (Differentiable Implies Continuous)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, then f is continuous at c.*

Go to proof.

Theorem 0.57.7 (Uniqueness of the Derivative)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, its derivative is unique.*

Go to proof. Go to proof.

Proposition 0.57.8 (Derivative of the Identity)**.** *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

Go to proof.

Proposition 0.57.9 (Derivative of a Constant)**.** *Let $k \in \mathbb{R}$ and let $f(x) = k$ for all $x \in \mathbb{R}$. Then $f'(c) = 0$ for every $c \in \mathbb{R}$.*

Go to proof.

0.58 One-Sided Derivatives

Definition 0.58.1 (Left-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I. We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}.$$

Definition 0.58.2 (Right-Hand Derivative)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I. We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Theorem 0.58.3 (Differentiability and One-Sided Derivatives)**.** *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$. Then f is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

0.59 Carathéodory and the Chain Rule

0.59.1 Carathéodory Characterization

Definition 0.59.1 (Carathéodory Factorization)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Theorem 0.59.2 (Carathéodory Characterization of Differentiability)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

(i) f is differentiable at c.

(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \textit{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. Go to proof. Go to proof.

Theorem 0.59.3 (Chain Rule)**.** *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c, and*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

Go to proof.

Theorem 0.59.4 (Leibniz Rule)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $x \in I$, and suppose that $f, g : I \to \mathbb{R}$ are n-times differentiable on a neighbourhood of x. Then fg is n-times differentiable at x, and*

$$(fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Go to proof.

Theorem 0.59.5 (Faa di Bruno's Formula)**.** *Let $I, J \subseteq \mathbb{R}$ be intervals, let $x \in I$, suppose $g : I \to J$ is n-times differentiable on a neighbourhood of x, and suppose $f : J \to \mathbb{R}$ is n-times differentiable on a neighbourhood of $g(x)$. Then $f \circ g$ is n-times differentiable at x, and*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^{n} m k_m = n} \frac{n!}{\prod_{m=1}^{n} k_m!\,(m!)^{k_m}}\, f^{(k)}(g(x)) \prod_{m=1}^{n} \left(g^{(m)}(x)\right)^{k_m},$$

where $k_m \geq 0$ and $k = \sum_{m=1}^{n} k_m$. Go to proof.

0.60 The Shape Questions

Definition 0.60.1 (Convex Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *convex* on I if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda) f(y).$$

Definition 0.60.2 (Concave Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *concave* on I if $-f$ is convex on I.

Definition 0.60.3 (Inflection Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be an interior point. The point c is an *inflection point* of f if f changes convexity at c: there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that f is convex on one side of c and concave on the other side.

0.60.1 Interior Extrema

Definition 0.60.4 (Relative Minimum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \le f(x) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(c) < f(x) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative minimum* at c.

Definition 0.60.5 (Relative Maximum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \le f(c) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(x) < f(c) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative maximum* at c.

Definition 0.60.6 (Relative Extremum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c.

Theorem 0.60.7 (Interior Extremum Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c, then*

$$f'(c) = 0.$$

Go to proof.

Theorem 0.60.8 (Necessary Condition for an Interior Extremum)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Go to proof.

Corollary 0.60.9 (Necessary Condition for an Extremum)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then*

$$f'(c) = 0 \quad \textit{or} \quad f'(c) \ \textit{does not exist.}$$

Go to proof.

0.61 The Mean Value Theorems

Theorem 0.61.1 (Rolle's Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

Go to proof.

Theorem 0.61.2 (Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Theorem 0.61.3 (Cauchy Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$\big(f(b) - f(a)\big)g'(c) = \big(g(b) - g(a)\big)f'(c).$$

If, in addition, $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Go to proof.

Corollary 0.61.4 (Derivative Bound Implies Lipschitz)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. If there exists $M \geq 0$ such that*

$$|f'(x)| \leq M \qquad \text{for every } x \in I,$$

then f is Lipschitz on I with constant M:

$$|f(x) - f(y)| \leq M|x - y| \qquad \text{for every } x, y \in I.$$

Go to proof.

0.62 Reading the Graph from f' and f''

0.62.1 First-Order Shape

Definition 0.62.1 (Increasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *increasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \leq f(c) \leq f(y)$.

Definition 0.62.2 (Decreasing at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *decreasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \geq f(c) \geq f(y)$.

Theorem 0.62.3 (Zero Derivative Implies Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = 0 \qquad \text{for all } x \in (a, b),$$

then f is constant on I. Go to proof.

Corollary 0.62.4 (Equal Derivatives Imply Difference Is Constant)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, and let $f, g : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = g'(x) \qquad \text{for all } x \in (a, b),$$

then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$. Go to proof.

Theorem 0.62.5 (Nondecreasing Iff Nonnegative Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nondecreasing on I if and only if*

$$f'(x) \geq 0 \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.62.6 (Nonincreasing Iff Nonpositive Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nonincreasing on I if and only if*

$$f'(x) \le 0 \qquad \text{for all } x \in I.$$

Go to proof.

Lemma 0.62.7 (Positive Derivative Implies Locally Increasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Lemma 0.62.8 (Negative Derivative Implies Locally Decreasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Theorem 0.62.9 (First Derivative Test: Relative Maximum)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \ge 0 \quad \text{for all } x \in (c - \delta, c), \qquad f'(x) \le 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative maximum at c. *Go to proof.*

Theorem 0.62.10 (First Derivative Test: Relative Minimum)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \le 0 \quad \text{for all } x \in (c - \delta, c), \qquad f'(x) \ge 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative minimum at c. *Go to proof.*

0.62.2 Second-Order Shape

0.62.3 Higher-Order Derivatives

Definition 0.62.11 (Second Derivative)**.** Let f be differentiable on a neighbourhood of c. If f' is differentiable at c, the *second derivative* of f at c is

$$f''(c) := (f')'(c).$$

Definition 0.62.12 (Higher Derivatives)**.** Define $f^{(0)} := f$ and $f^{(1)} := f'$. For $n \ge 2$, if $f^{(n-1)}$ is differentiable at c, define

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

0.62.4 Second-Order Shape

Theorem 0.62.13 (Second-Derivative Convexity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*

$$f''(x) \geq 0 \qquad \text{for all } x \in I,$$

then f is convex on I. *Go to proof.*

Theorem 0.62.14 (Second-Derivative Concavity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*

$$f''(x) \leq 0 \qquad \text{for all } x \in I,$$

then f is concave on I. *Go to proof.*

Theorem 0.62.15 (Second Derivative Test)**.** *Let f be twice differentiable in a neighbourhood of c, and suppose $f'(c) = 0$.*

- *If $f''(c) > 0$, then f has a strict relative minimum at c.*
- *If $f''(c) < 0$, then f has a strict relative maximum at c.*

Go to proof.

Proposition 0.62.16 (Inflection Point Necessary Condition)**.** *Let f have an inflection point at c. If f'' exists on a neighbourhood of c and is continuous at c, then*

$$f''(c) = 0.$$

Go to proof.

0.62.5 Darboux's Theorem

Theorem 0.62.17 (Darboux's Theorem)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on I. If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = k.$$

Go to proof.

0.62.6 Smoothness Classes

Definition 0.62.18 (Continuously Differentiable; Class C^1)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *continuously differentiable* on I, written $f \in C^1(I)$, if f is differentiable at every point of I and the derivative $f' : I \to \mathbb{R}$ is continuous on I.

Definition 0.62.19 (Class C^k)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $k \in \mathbb{N}$, and let $f : I \to \mathbb{R}$. The function f is of *class C^k* on I, written $f \in C^k(I)$, if the derivatives $f^{(0)}, f^{(1)}, \ldots, f^{(k)}$ exist on I and $f^{(k)}$ is continuous on I. By convention, $C^0(I)$ is the class of continuous functions on I.

Definition 0.62.20 (Smooth Function; Class C^∞)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *smooth*, or of *class C^∞*, written $f \in C^\infty(I)$, if $f \in C^k(I)$ for every $k \in \mathbb{N}$. Equivalently,

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

Definition 0.62.21 (Real-Analytic Function; Class C^ω)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *real-analytic*, or of *class C^ω*, written $f \in C^\omega(I)$, if for every $a \in I$ there exists $r > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x - a)^n \qquad \text{for every } x \in (a - r, a + r) \cap I.$$

Theorem 0.62.22 (The Smoothness Tower)**.** *For every nondegenerate interval I, the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

hold, and each displayed inclusion is strict. Go to proof.

Definition 0.62.23 (Lipschitz-Derivative Class $C^{1,1}$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is of *class* $C^{1,1}$ on I, written $f \in C^{1,1}(I)$, if $f \in C^1(I)$ and f' is Lipschitz on I: there exists $L \geq 0$ such that

$$|f'(x) - f'(y)| \leq L|x - y| \qquad \text{for all } x, y \in I.$$

Theorem 0.62.24 (Placement of $C^{1,1}$)**.** *For every nondegenerate interval I,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

and $C^{1,1}(I)$ is not contained in $C^2(I)$. If K is a compact interval, then

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

and the first inclusion is strict. Go to proof.

Corollary 0.62.25 (Bounded Second Derivative Implies $C^{1,1}$)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f \in C^2(I)$. If $|f''(x)| \leq M$ for all $x \in I$, then $f \in C^{1,1}(I)$, and f' is Lipschitz with constant M. Go to proof.*

0.63 The Algebra of Derivatives

0.63.1 From Characterization to Computation

Theorem 0.63.1 (Constant Multiple Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $\alpha \in \mathbb{R}$. If f is differentiable at c, then αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof.

Theorem 0.63.2 (Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then $f + g$ is differentiable at c, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Theorem 0.63.3 (Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Theorem 0.63.4 (Quotient Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

0.63.2 Closure

Corollary 0.63.5 (Finite Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable at c, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i \right)' (c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

Corollary 0.63.6 (Extended Product Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\prod_{i=1}^{n} f_i$ is differentiable at c, and*

$$\left(\prod_{i=1}^{n} f_i \right)' (c) = \sum_{k=1}^{n} f_k'(c) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(c).$$

Go to proof.

Corollary 0.63.7 (Power Rule for Integer Powers of a Function). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $n \in \mathbb{N}$. If f is differentiable at c, then f^n is differentiable at c, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Go to proof.

Theorem 0.63.8 (Finite Linear Combination Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i \right)' (c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

0.63.3 Global Forms on an Interval

Theorem 0.63.9 (Constant Multiple Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I, then αf is differentiable on I, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.63.10 (Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then $f + g$ is differentiable on I, and*

$$(f + g)'(x) = f'(x) + g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.63.11 (Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then fg is differentiable on I, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.63.12 (Quotient Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \text{for all } x \in I.$$

Go to proof.

Corollary 0.63.13 (Finite Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable on I, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Corollary 0.63.14 (Extended Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \ldots, f_n : I \to \mathbb{R}$. If each f_i is differentiable on I, then $\prod_{i=1}^{n} f_i$ is differentiable on I, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(x) = \sum_{k=1}^{n} f_k'(x) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(x) \qquad \text{for all } x \in I.$$

Go to proof.

Corollary 0.63.15 (Power Rule for Integer Powers of a Function on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I, then f^n is differentiable on I, and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.63.16 (Finite Linear Combination Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \ldots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

0.63.4 Inverting the Operation

Theorem 0.63.17 (Inverse Function Theorem, One Variable). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$ be of class C^1, and let $c \in I$ satisfy $f'(c) \neq 0$. Then there exist open intervals $U \subseteq I$ and $V \subseteq \mathbb{R}$, with $c \in U$ and $f(c) \in V$, such that:*

1. *$f|_U : U \to V$ is a bijection;*
2. *the inverse $g = (f|_U)^{-1} : V \to U$ is of class C^1;*
3. *for every $y \in V$,*
$$g'(y) = \frac{1}{f'(g(y))}.$$

Go to proof.

Corollary 0.63.18 (Derivative of the Inverse Function)**.** *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse g satisfies*

$$g' = \frac{1}{f' \circ g}$$

on V. More globally, if $f : I \to J$ is a C^1 bijection with $f'(x) \neq 0$ for every $x \in I$, then $f^{-1} : J \to I$ is C^1 and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \qquad \textit{for every } y \in J.$$

Go to proof.

0.63.5 Indeterminate Forms: L'Hôpital's Rule

Theorem 0.63.19 (L'Hôpital's Rule, 0/0 Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x \to a^+} f(x) = \lim_{x \to a^+} g(x) = 0 \qquad \textit{and} \qquad \lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.

Theorem 0.63.20 (L'Hôpital's Rule, ∞/∞ Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x \to a^+} |g(x)| = +\infty \qquad \textit{and} \qquad \lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

No separate hypothesis on $\lim_{x \to a^+} f(x)$ is required. Go to proof.

0.64 Taylor: Construction and Its Error

Definition 0.64.1 (Taylor Polynomial at a Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \leq k \leq n$. The **Taylor polynomial of order** n for f at a is

$$T_{n,a} f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

Definition 0.64.2 (Taylor Remainder)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have a Taylor polynomial $T_{n,a} f$. The **Taylor remainder of order** n at a is the function $R_{n,a} f : I \to \mathbb{R}$ defined by

$$R_{n,a} f(x) := f(x) - T_{n,a} f(x).$$

Definition 0.64.3 (Maclaurin Polynomial)**.** Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \leq k \leq n$. The **Maclaurin polynomial of order** n for f is

$$T_{n,0} f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k.$$

Theorem 0.64.4 (Taylor's Theorem with Lagrange Remainder). *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \to \mathbb{R}$. Suppose that $f, f', \dots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b). Then, for every $x \in (a, b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k + \frac{f^{(n+1)}(c)}{(n + 1)!}(x - a)^{n+1}.$$

Go to proof.

Corollary 0.64.5 (Taylor Expansion with Peano Remainder). *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order n in a neighbourhood of a. If $f^{(n)}$ is continuous at a, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k + o\big((x - a)^n\big) \qquad \text{as } x \to a.$$

Go to proof.

Corollary 0.64.6 (First-Order Peano Remainder). *Let f be differentiable at c. Then, as $h \to 0$,*

$$f(c + h) = f(c) + f'(c)h + o(h).$$

Go to proof.

Proposition 0.64.7 (The Flat Function: Smooth but Not Analytic). *Define $f : \mathbb{R} \to \mathbb{R}$ by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then:

1. *$f \in C^\infty(\mathbb{R})$;*
2. *$f^{(n)}(0) = 0$ for every $n \geq 0$;*
3. *the Maclaurin series of f is identically 0, so $f \notin C^\omega$ on any neighbourhood of 0.*

In particular, $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \varnothing$. *Go to proof.*

0.64.1 The Differential

Definition 0.64.8 (Differentiability by a Differential). Let f be defined on a neighbourhood of $c \in \mathbb{R}$. The function f is *differentiable at c in the differential sense* if there exists a linear map $L : \mathbb{R} \to \mathbb{R}$ such that

$$f(c + h) - f(c) = L(h) + o(h) \qquad \text{as } h \to 0.$$

In that case the *differential* of f at c is $df_c := L$.

Theorem 0.64.9 (Differential and Derivative Agree). *Let f be defined on a neighbourhood of $c \in \mathbb{R}$. Then f is differentiable at c in the ordinary derivative sense if and only if it is differentiable at c in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

Go to proof.

Theorem 0.64.10 (Uniqueness of the Differential). *If $L_1, L_2 : \mathbb{R} \to \mathbb{R}$ are linear maps and*

$$f(c + h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \qquad \text{as } h \to 0,$$

then $L_1 = L_2$. *Go to proof.*

Theorem 0.64.11 (Differential Continuity Criterion)**.** *If f is differentiable at c in the differential sense, then f is continuous at c.* *Go to proof.*

Theorem 0.64.12 (Chain Rule for Differentials)**.** *If f is differentiable at c and g is differentiable at $f(c)$, then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Go to proof.

Theorem 0.64.13 (Linearity of the Differential)**.** *If f and g are differentiable at c and $\alpha, \beta \in \mathbb{R}$, then*

$$d(\alpha f + \beta g)_c = \alpha\, df_c + \beta\, dg_c.$$

Go to proof.

Proofs

Classical Analysis

Integration

0.65 Partitions and Tags

The Geometric Problem

Definition 0.65.1 (Partition of a Compact Interval)**.** A *partition* of $[a, b]$ is a finite increasing list

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Equivalently,

$$\begin{aligned}\text{Partition}(P, a, b) \iff\ & \exists n \in \mathbb{N}\, \exists x_0, \ldots, x_n \in \mathbb{R} \\ & \big(P = \{x_0, \ldots, x_n\} \wedge x_0 = a \wedge x_n = b\big) \\ & \wedge \forall i \, \big(1 \le i \le n \Rightarrow x_{i-1} < x_i\big).\end{aligned}$$

Definition 0.65.2 (Subinterval Width)**.** The i-th slab has width

$$\Delta x_i = x_i - x_{i-1}.$$

Definition 0.65.3 (Mesh of a Partition)**.** The *mesh* or *norm* of P is the width of its fattest slab:

$$\|P\| = \max_{1 \le i \le n} \Delta x_i.$$

Definition 0.65.4 (Tagged Partition)**.** A *tagged partition* is a partition $P = \{x_0, \ldots, x_n\}$ together with points $\xi_i \in [x_{i-1}, x_i]$. The tag is the point where the height $f(\xi_i)$ is sampled.

Definition 0.65.5 (Refinement of a Partition)**.** A partition P' *refines* P when $P \subseteq P'$. Refinement adds cut-points; it does not move or delete existing cut-points.

Lemma 0.65.6 (Common Refinement of Two Partitions)**.** *Any two partitions P and P' of $[a, b]$ have a common refinement: order the finite set $P \cup P'$ increasingly.*

0.66 The Cauchy Integral

Geometric Intuition

Definition 0.66.1 (Left-Endpoint Cauchy Sum)**.** For $f : [a, b] \to \mathbb{R}$ and $P = \{a = x_0 < \cdots < x_n = b\}$, define

$$C(f, P) = \sum_{i=1}^{n} f(x_{i-1})\, \Delta x_i.$$

Definition 0.66.2 (Cauchy Integral)**.** The function f is *Cauchy-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall P\ \big(\|P\| < \delta \Rightarrow |C(f, P) - L| < \varepsilon\big).$$

The value L is denoted $\int_a^b f(x)\, dx$.

Proposition 0.66.3 (Cauchy Integral of a Constant). *If $f(x) = c$ on $[a, b]$, then $\int_a^b c\,dx = c(b-a)$.*

Theorem 0.66.4 (Linearity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem 0.66.5 (Monotonicity of the Cauchy Integral). *If f and g are Cauchy-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\,dx \le \int_a^b g\,dx.$$

Corollary 0.66.6 (Bounds for the Cauchy Integral). *If f is continuous on $[a, b]$, $m = \min_{[a,b]} f$, and $M = \max_{[a,b]} f$, then*

$$m(b-a) \le \int_a^b f\,dx \le M(b-a).$$

Theorem 0.66.7 (Triangle Inequality for the Cauchy Integral). *If f and $|f|$ are Cauchy-integrable on $[a, b]$, then*

$$\left| \int_a^b f\,dx \right| \le \int_a^b |f|\,dx.$$

Theorem 0.66.8 (Interval Additivity for the Cauchy Integral). *If $c \in [a, b]$ and f is Cauchy-integrable on the relevant intervals, then*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Definition 0.66.9 (Oscillation on an Interval). For bounded f on an interval I, define

$$\omega_f(I) = \sup_I f - \inf_I f.$$

Oscillation is the error controller: if $\omega_f(I)$ is small, every sample height on I is close to every other sample height.

Theorem 0.66.10 (Continuous Functions Are Cauchy-Integrable). *Every continuous function $f : [a, b] \to \mathbb{R}$ is Cauchy-integrable.*

Theorem 0.66.11 (Tag Independence for Continuous Functions). *If f is continuous on $[a, b]$, then every choice of tags gives the same limit as the left-endpoint sums as $\|P\| \to 0$.*

0.67 The Riemann Integral

Geometric Intuition

Definition 0.67.1 (Riemann Sum). For a tagged partition (P, ξ), define

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

Definition 0.67.2 (Riemann Integral). A function $f : [a, b] \to \mathbb{R}$ is *Riemann-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall (P, \xi)\ \big(\|P\| < \delta \Rightarrow |S(f, P, \xi) - L| < \varepsilon\big).$$

Theorem 0.67.3 (Continuous Functions Are Riemann-Integrable). *Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.*

Theorem 0.67.4 (Linearity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\, dx = \alpha \int_a^b f\, dx + \beta \int_a^b g\, dx.$$

Theorem 0.67.5 (Monotonicity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a, b]$ and $f(x) \leq g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\, dx \leq \int_a^b g\, dx.$$

Theorem 0.67.6 (Triangle Inequality for the Riemann Integral)**.** *If f is Riemann-integrable on $[a, b]$, then $|f|$ is Riemann-integrable and*

$$\left| \int_a^b f\, dx \right| \leq \int_a^b |f|\, dx.$$

Theorem 0.67.7 (Interval Additivity for the Riemann Integral)**.** *If $c \in [a, b]$, then f is Riemann-integrable on $[a, b]$ if and only if it is Riemann-integrable on $[a, c]$ and $[c, b]$, and in that case*

$$\int_a^b f\, dx = \int_a^c f\, dx + \int_c^b f\, dx.$$

Theorem 0.67.8 (Cauchy Criterion for Riemann Integrability)**.** *A bounded function is Riemann-integrable if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that any two tagged partitions (P, ξ) and (Q, η) with $\|P\|, \|Q\| < \delta$ have*

$$|S(f, P, \xi) - S(f, Q, \eta)| < \varepsilon.$$

0.68 The Darboux Integral

Geometric Intuition

Definition 0.68.1 (Lower and Upper Darboux Sums)**.** For bounded f on $[a, b]$, set

$$m_i = \inf_{[x_{i-1}, x_i]} f, \qquad M_i = \sup_{[x_{i-1}, x_i]} f,$$

and define

$$L(f, P) = \sum_i m_i \Delta x_i, \qquad U(f, P) = \sum_i M_i \Delta x_i.$$

Lemma 0.68.2 (Darboux Sums Under Refinement)**.** *If P' refines P, then*

$$L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P).$$

Definition 0.68.3 (Darboux Integrability)**.** The lower and upper integrals are

$$\underline{I} = \sup_P L(f, P), \qquad \overline{I} = \inf_P U(f, P).$$

The function f is *Darboux-integrable* if $\underline{I} = \overline{I}$.

Theorem 0.68.4 (Darboux Criterion)**.** *A bounded function f is Darboux-integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that*

$$U(f, P) - L(f, P) < \varepsilon.$$

Theorem 0.68.5 (Equivalence of Riemann and Darboux Integrability)**.** *For bounded functions on $[a, b]$, Riemann integrability and Darboux integrability are equivalent, and the integral values agree.*

Theorem 0.68.6 (Continuous Functions Are Darboux-Integrable). *Every continuous function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem 0.68.7 (Monotone Functions Are Darboux-Integrable). *Every monotone function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem 0.68.8 (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable). *If $f : [a, b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Darboux-integrable.*

Theorem 0.68.9 (Sums and Scalar Multiples of Darboux-Integrable Functions). *If f and g are Darboux-integrable on $[a, b]$, then $\alpha f + \beta g$ is Darboux-integrable for all $\alpha, \beta \in \mathbb{R}$.*

Theorem 0.68.10 (Products of Darboux-Integrable Functions). *If f and g are Darboux-integrable on $[a, b]$, then fg is Darboux-integrable on $[a, b]$.*

Theorem 0.68.11 (Absolute Values of Darboux-Integrable Functions). *If f is Darboux-integrable on $[a, b]$, then $|f|$ is Darboux-integrable on $[a, b]$.*

Theorem 0.68.12 (Continuous Images of Darboux-Integrable Functions). *If f is Darboux-integrable on $[a, b]$, φ is continuous on an interval containing $f([a, b])$, and f is bounded, then $\varphi \circ f$ is Darboux-integrable on $[a, b]$.*

0.69 The Riemann–Stieltjes Integral

Geometric Intuition

Definition 0.69.1 (Total Variation). The total variation of α on $[a, b]$ is

$$V_a^b(\alpha) = \sup_P \sum_i |\alpha(x_i) - \alpha(x_{i-1})|.$$

Definition 0.69.2 (Bounded Variation). The function α has *bounded variation* on $[a, b]$ if

$$V_a^b(\alpha) < \infty.$$

Geometrically, this says the ruler has finite total up-and-down travel.

Proposition 0.69.3 (Monotone Functions Have Bounded Variation). *If α is monotone on $[a, b]$, then α has bounded variation.*

Definition 0.69.4 (Riemann–Stieltjes Integral). The Riemann–Stieltjes sum is

$$\sum_i f(\xi_i)\big(\alpha(x_i) - \alpha(x_{i-1})\big).$$

When these sums have a tag-independent limit as $\|P\| \to 0$, the limit is denoted $\int_a^b f\, d\alpha$.

Theorem 0.69.5 (Continuous Integrand and BV Integrator). *If f is continuous on $[a, b]$ and α has bounded variation, then $\int_a^b f\, d\alpha$ exists.*

Theorem 0.69.6 (Bilinearity of the Riemann–Stieltjes Integral). *Whenever the displayed integrals exist,*

$$\int_a^b (\lambda f + \mu g)\, d\alpha = \lambda \int_a^b f\, d\alpha + \mu \int_a^b g\, d\alpha$$

and

$$\int_a^b f\, d(\lambda\alpha + \mu\beta) = \lambda \int_a^b f\, d\alpha + \mu \int_a^b f\, d\beta.$$

Theorem 0.69.7 (Interval Additivity for the Riemann–Stieltjes Integral). *If $c \in [a, b]$ and the relevant Riemann–Stieltjes integrals exist, then*

$$\int_a^b f\,d\alpha = \int_a^c f\,d\alpha + \int_c^b f\,d\alpha.$$

Theorem 0.69.8 (Riemann–Stieltjes Integration by Parts). *If f and α are such that one of $\int_a^b f\,d\alpha$ and $\int_a^b \alpha\,df$ exists, then the other exists under the same Riemann–Stieltjes convention, and*

$$\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a).$$

Theorem 0.69.9 (Reduction to Riemann Integration for a C^1 Integrator). *If $\alpha \in C^1([a, b])$ and f is Riemann-integrable on $[a, b]$, then*

$$\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx.$$

Theorem 0.69.10 (Step Integrators Give Finite Sums). *If α is constant except for finitely many jumps at points c_k and f is continuous at each c_k, then*

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

0.70 The Henstock–Kurzweil Integral

Where This Sits

Geometric Intuition

Definition 0.70.1 (Gauge on an Interval). A *gauge* on $[a, b]$ is a function

$$\delta : [a, b] \to (0, \infty).$$

Definition 0.70.2 (δ-Fine Tagged Partition). A tagged partition (P, ξ) is *δ-fine* if

$$[x_{i-1}, x_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Definition 0.70.3 (Henstock–Kurzweil Integral). A function $f : [a, b] \to \mathbb{R}$ is *Henstock–Kurzweil integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a, b] \to (0, \infty)\ \forall (P, \xi)\ \big((P, \xi) \text{ is } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon\big).$$

The value is denoted (HK) $\int_a^b f$.

Lemma 0.70.4 (Cousin's Lemma). *For every gauge δ on $[a, b]$, there exists a δ-fine tagged partition of $[a, b]$.*

Theorem 0.70.5 (Riemann Integrable Implies Henstock–Kurzweil Integrable). *If f is Riemann-integrable on $[a, b]$, then f is Henstock–Kurzweil integrable on $[a, b]$, and the two integrals have the same value.*

Lemma 0.70.6 (Straddle Lemma). *If F is differentiable at ξ with $F'(\xi) = f(\xi)$, then for every $\varepsilon > 0$ there is $\delta(\xi) > 0$ such that whenever $u \le \xi \le v$, $u, v \in (\xi - \delta(\xi), \xi + \delta(\xi))$,*

$$|F(v) - F(u) - f(\xi)(v - u)| \le \varepsilon(v - u).$$

Theorem 0.70.7 (Fundamental Theorem for the Henstock–Kurzweil Integral). *If $F : [a, b] \to \mathbb{R}$ is differentiable at every point and $F' = f$, then f is Henstock–Kurzweil integrable and*

$$(\mathrm{HK}) \int_a^b f = F(b) - F(a).$$

Theorem 0.70.8 (Continuous Functions Are Henstock–Kurzweil Integrable). *Every continuous function $f : [a, b] \to \mathbb{R}$ is Henstock–Kurzweil integrable.*

0.71 The McShane Integral

Where This Sits

Geometric Intuition

Definition 0.71.1 (McShane-Fine Tagged Partition)**.** Let $\delta : [a, b] \to (0, \infty)$ be a gauge. A finite tagged partition $\{([u_i, v_i], \xi_i)\}_{i=1}^{n}$ is *McShane δ-fine* if the intervals $[u_i, v_i]$ are non-overlapping, cover $[a, b]$, each $\xi_i \in [a, b]$, and

$$[u_i, v_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Unlike an HK tagged partition, the tag ξ_i is not required to lie in $[u_i, v_i]$.

Definition 0.71.2 (McShane Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *McShane integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a, b] \to (0, \infty)\ \forall (P, \xi)\ \big((P, \xi) \text{ is McShane } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon\big),$$

where

$$S(f, P, \xi) = \sum_i f(\xi_i)(v_i - u_i).$$

Theorem 0.71.3 (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable)**.** *On a compact interval,*

$$\mathcal{R}[a, b] \subseteq \mathcal{M}[a, b] \subseteq \mathcal{HK}[a, b],$$

and the integral values agree whenever a function belongs to the smaller class.

Theorem 0.71.4 (McShane Integrability Equals Lebesgue Integrability)**.** *For real-valued functions on a compact interval,*

$$\mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a, b] = \mathcal{L}[a, b],$$

and the McShane and Lebesgue integral values agree.

0.72 Measure Zero and the Lebesgue Criterion

Geometric Intuition

Definition 0.72.1 (Measure Zero)**.** A set $E \subseteq \mathbb{R}$ has *measure zero* if for every $\varepsilon > 0$ there are open intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \qquad \sum_{k=1}^{\infty} |I_k| < \varepsilon.$$

Definition 0.72.2 (Point Oscillation)**.** For bounded f on $[a, b]$, define

$$\omega_f(x) = \inf_{\delta > 0} \omega_f\big((x - \delta, x + \delta) \cap [a, b]\big).$$

Then f is continuous at x exactly when $\omega_f(x) = 0$.

Theorem 0.72.3 (Lebesgue Criterion for Riemann Integrability)**.** *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if its discontinuity set*

$$D = \{x \in [a, b] : \omega_f(x) > 0\}$$

has measure zero.

Proofs